

Unemployment Dynamics in the South-Central United States: Single-State and Cross-State Time Series Analysis

1. Introduction

This report focuses on unemployment dynamics in seven South-Central U.S. states: Texas, Oklahoma, Arkansas, Kansas, Missouri, Nebraska, and New Mexico. The main question is how unemployment behaves within each state and whether unemployment changes in one state are related to changes in others. Since these states are located in the same general region but differ in economic structure and labor-market size, they provide a useful setting for both individual-state and regional analysis.

The report uses both single-state and cross-state methods. In the single-state analysis, We examine both non-seasonally adjusted (NSA) and seasonally adjusted (SA) unemployment series so that we can compare seasonal patterns with the underlying trend. This part uses STL decomposition, differencing, ACF/PACF, ARIMA and SARIMA models, and forecasting. In the cross-state analysis, we use VAR, IRFs, FEVD, and multistate forecasts to see whether unemployment shocks are transmitted across states. Together, these methods allow the report to study not only each state's own unemployment pattern but also the broader regional relationships among them.

2. Regional Background

2.1 Why the South-Central Region Matters

The South-Central region is a useful group to study because these states are close to each other, share some common economic conditions, and are still different enough to make comparison meaningful. They are affected by the same national recessions and recoveries, but they do not all respond in the same way. This makes the region a good setting for asking two related questions: how unemployment behaves within each state, and whether unemployment movements are linked across states.

Another reason this region matters is that unemployment changes in one state may not remain entirely local. Because these states share economic ties and are affected by common business-cycle conditions, labor-market shocks may spill over across state borders. A slowdown in one major state may influence neighboring states through supply chains, transportation networks, commodity markets, or reduced regional demand. For this reason, the South-Central region is a useful case for studying not only how unemployment evolves within individual states, but also whether unemployment rates move together across states over time.

2.2 Economic Structure of the Region

Although the seven states belong to the same broad region, their economic structures differ substantially. Texas has the largest and most diversified economy in the group, with major roles for energy, manufacturing, logistics, technology, and a wide range of service industries. Oklahoma is smaller and more concentrated, with stronger dependence on energy, agriculture, and transportation-related activity. Arkansas has a different profile, relying more on retail, food processing, logistics, and agriculture, with large employers in distribution and consumer goods helping support labor demand.

Kansas combines agriculture with aviation manufacturing, broader manufacturing activity, and some energy-related exposure. Missouri has a more mixed economy, with manufacturing, transportation, healthcare, and service industries all playing important roles. Nebraska is strongly tied to agriculture, livestock, and food production, which contributes to its relatively stable labor-market behavior. New Mexico has a more specialized structure, with stronger dependence on energy, government-related activity, and tourism, making it somewhat more sensitive to cyclical and seasonal shifts.

These differences matter because unemployment is shaped not only by national recessions but also by the kinds of industries that dominate a state economy. States with large, diversified labor markets may be better able to absorb shocks, while states with narrower industrial bases may respond more sharply when key sectors weaken.

2.3 Why Unemployment May Differ Across States

The variation in industrial structure across these states helps explain why unemployment may follow different paths even when all states are exposed to the same broad national shocks. More diversified economies, especially larger ones such as Texas, may show greater resilience because weakness in one sector can be partly offset by strength in others. By contrast, states that rely more heavily on a narrower group of industries may be more cyclical and more sensitive to commodity prices, trade conditions, or sector-specific downturns.

Seasonality may also differ across states. In states where agriculture, tourism, or related activities are more important, unemployment may show stronger recurring within-year movement. This helps explain why non-seasonally adjusted data remain useful in the report: the raw series can

preserve patterns that are economically meaningful rather than just statistical noise. At the same time, the seasonally adjusted series are useful for identifying the underlying labor-market trend once recurring monthly fluctuations are removed.

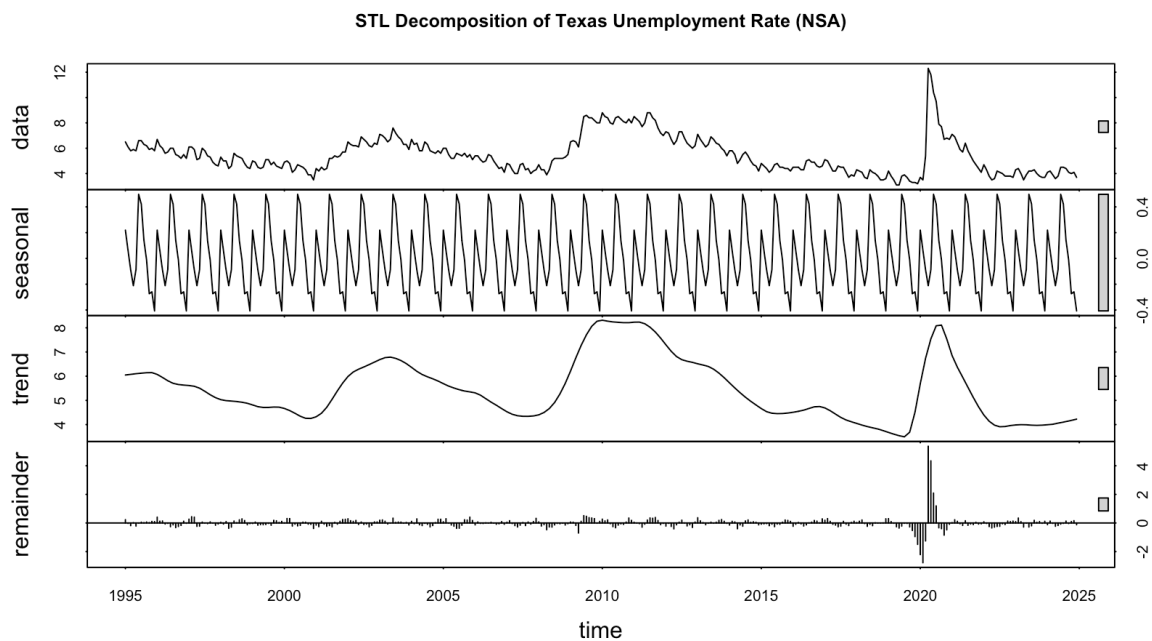
State size also matters for cross-state transmission. Larger states may have greater regional influence because their labor markets are more deeply connected to surrounding states through production networks, trade, transportation, and migration. Smaller states, in contrast, may be more likely to absorb outside shocks rather than transmit them. For this reason, differences in unemployment across the South-Central region should be understood as the result of both state-specific economic structure and broader regional linkage. This regional background helps explain why the report uses both single-state and cross-state methods in the sections that follow.

3. Individual State Analyses

3.1 Texas

Texas is the largest economy in this group, so its unemployment series is useful not only because of its own size, but also because it reflects how a large and diversified labor market absorbs shocks over time. Even before looking at the models, the broad pattern in the data is already quite clear. Texas unemployment falls during the late 1990s, rises again in the early 2000s, stays elevated for a fairly long period around the Great Recession, and then shows an extremely sharp spike in 2020 before settling back into a lower range afterward. The uploaded results describe this pattern in both the raw and adjusted data, and they also show that the pandemic shock is visually and statistically the most unusual part of the whole sample.

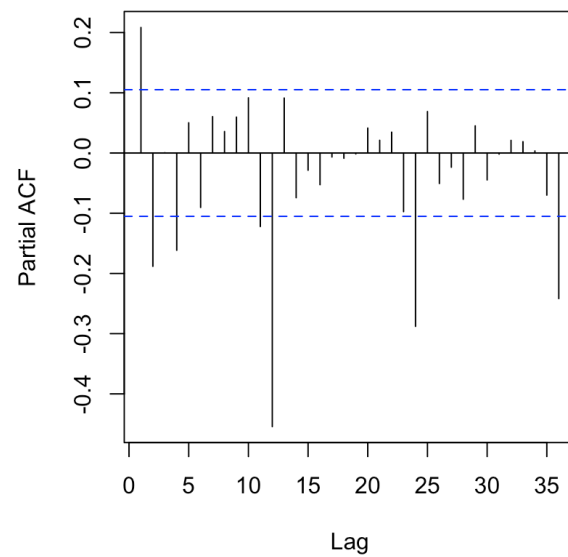
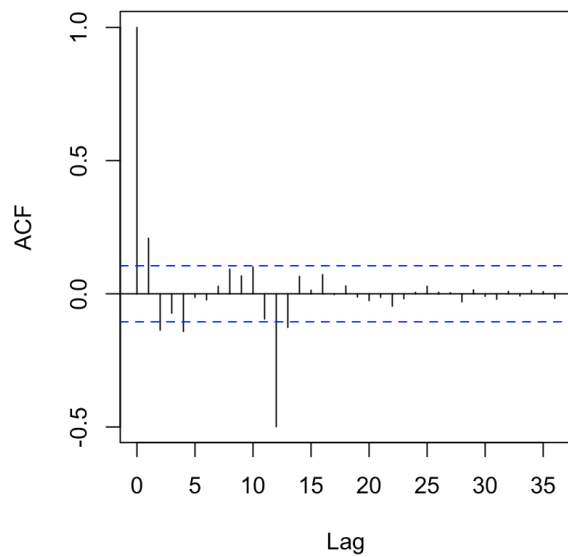
The non-seasonally adjusted series is the better place to start if the goal is to understand the full structure of the data rather than only the cleaned underlying trend. In the Texas NSA series, two things appear at the same time: there is a broad cyclical movement over the long run, but there is also a regular within-year seasonal pattern that repeats through the sample. The STL decomposition in the uploaded file makes that point quite clearly. The trend component moves with the major changes in the Texas labor market, the seasonal component remains fairly stable, and the remainder becomes unusually large during the pandemic period. That combination is important, because it means the raw series is not just noisy. It has structure, and that structure needs to be modeled explicitly.



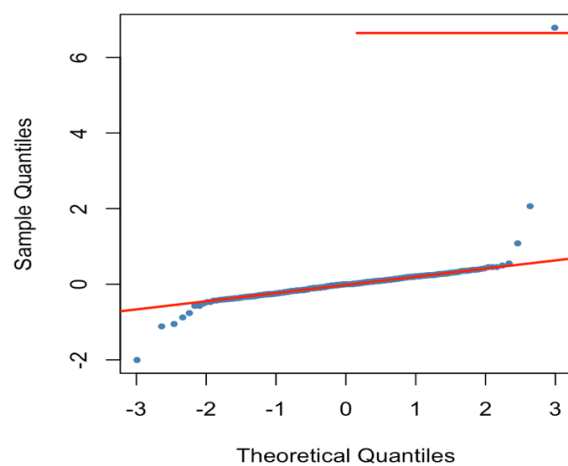
The differencing results also fit that story. Seasonal differencing helps remove the repeated annual movement, but it does not fully solve the persistence in the series. On the other hand, ordinary first differencing reduces the low-frequency trend but still leaves visible annual dependence. In other words, each transformation solves only part of the problem. Once both a

first difference and a seasonal difference at lag 12 are applied, the working series becomes much more stable and fluctuates around zero in a way that is more consistent with stationarity, aside from the obvious disruption around 2020. The uploaded identification results then support SARIMA(1,1,1)(0,1,1)[12] as the fitted model for the Texas NSA series.

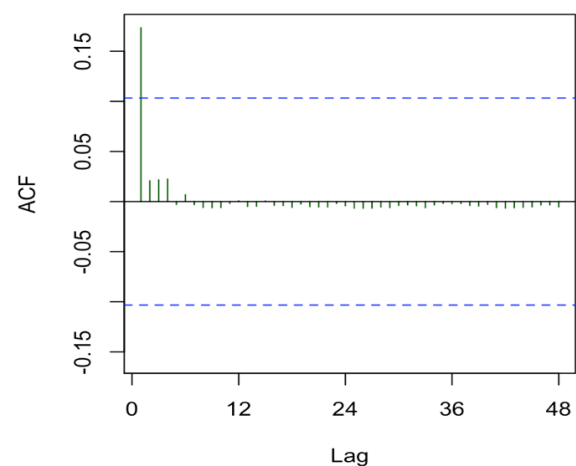
Texas NSA First + Seasonal Difference: ACF **Texas NSA First + Seasonal Difference: PAC**



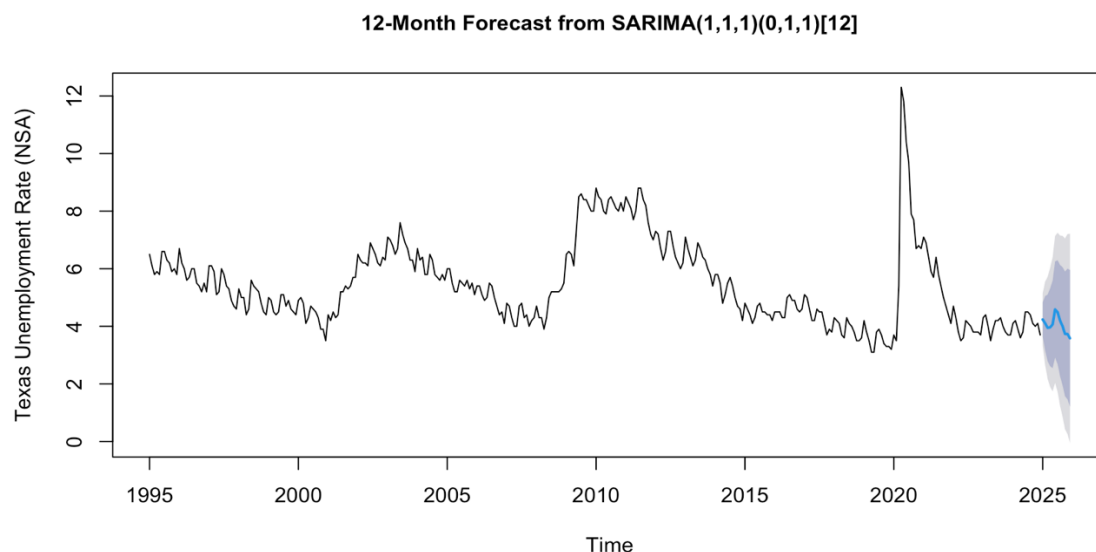
5. Q-Q Plot of Texas NSA Residuals



6. ACF of Residuals² (ARCH check)



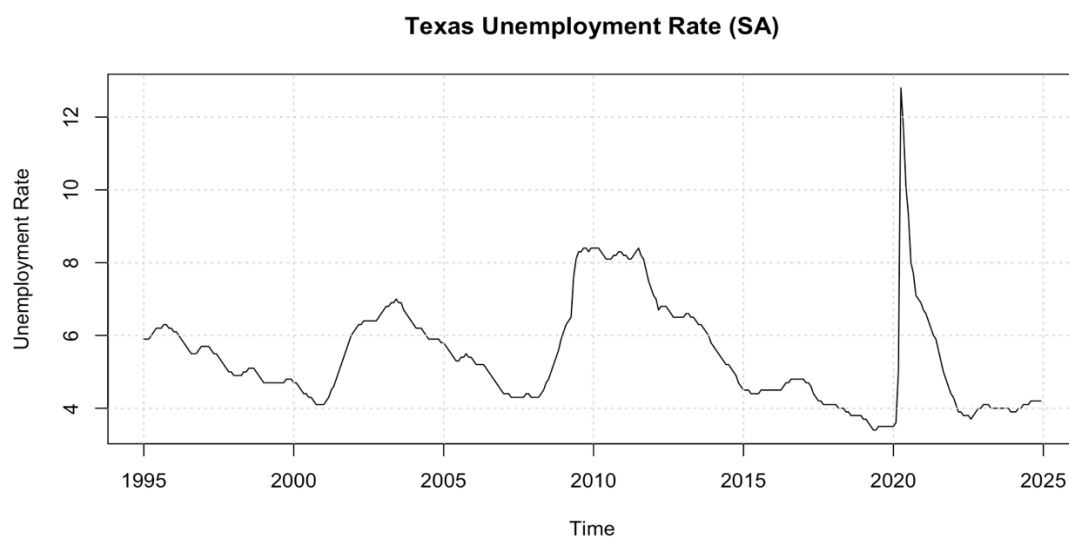
What makes this result convincing is not just the model label itself, but the way the diagnostics line up with it. The residual checks in the uploaded file suggest that the model does a decent job capturing the ordinary short-run movement in the data. Most of the residual autocorrelations stay within the confidence bounds, and the residual-squared ACF gives only weak evidence of strong ARCH-type clustering. That said, the model is clearly not perfect. The Q-Q plot still shows tail problems, which is not surprising, because a linear time-series model is not really designed to “explain away” something as extreme as the pandemic shock. So the most reasonable interpretation is that the model works well for ordinary months, but less well during rare crisis periods.



The 12-month forecast for the Texas NSA series is fairly restrained. The central path remains in the low- to mid-4% range, and the forecast fan widens gradually with the horizon. Those widening matters. It means the model is not claiming unusual precision; instead, it suggests that Texas is likely to remain near its recent unemployment range, while uncertainty naturally increases farther into the future. So the main message is not that Texas is improving dramatically

or weakening dramatically, but that the short-run outlook looks broadly stable once the exceptional 2020 shock is treated as history rather than a new baseline.

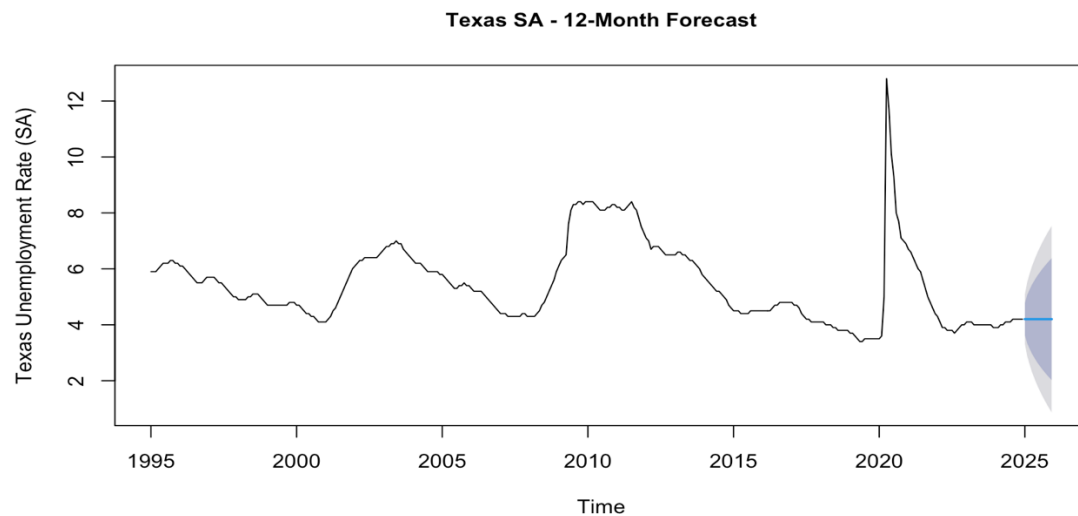
The seasonally adjusted Texas series tells the same general economic story, but in a cleaner form. Once the regular monthly seasonal pattern is removed, the remaining path is easier to read as an underlying labor-market cycle rather than a mixture of cycle and calendar effects. In the uploaded results, the SA series still shows the early-2000s rise, the higher unemployment period around 2009–2012, the dramatic jump in 2020, and the later return to a lower range. But compared with the NSA series, the path is smoother and easier to interpret. That is exactly why it is useful to keep both versions of the data in the report. The NSA series shows what households and firms actually experienced month by month, while the SA series makes it easier to discuss the underlying direction of the labor market.



Because the seasonal part has already been removed, the SA series does not need a seasonal ARIMA structure. The uploaded results show that one ordinary first difference is enough to

make the working series approximately stationary, and the chosen model is ARIMA(1,1,1). That is a simpler specification than in the NSA case, and that difference is not just a technical detail. It reflects the fact that seasonal adjustment has already done some of the work that otherwise would have to be done inside the model. Put differently, once the regular annual pattern is filtered out, the remaining unemployment dynamics in Texas can be described more parsimoniously.

The residual results for the SA model look broadly acceptable. Residual autocorrelation is limited, and the main distortion again comes from the pandemic outlier rather than from a general failure of the model. So the SA model seems to capture the ordinary trend dynamics reasonably well, even though it still cannot make the crisis period look “normal.” The forecast from the SA model is also easy to interpret: it is relatively flat and centered around the low-4% range. Compared with the NSA forecast, it looks smoother because the monthly seasonal variation has already been removed. Substantively, though, the conclusion is almost the same. Texas does not appear to be moving into a new period of sharply worsening unemployment over the next year.



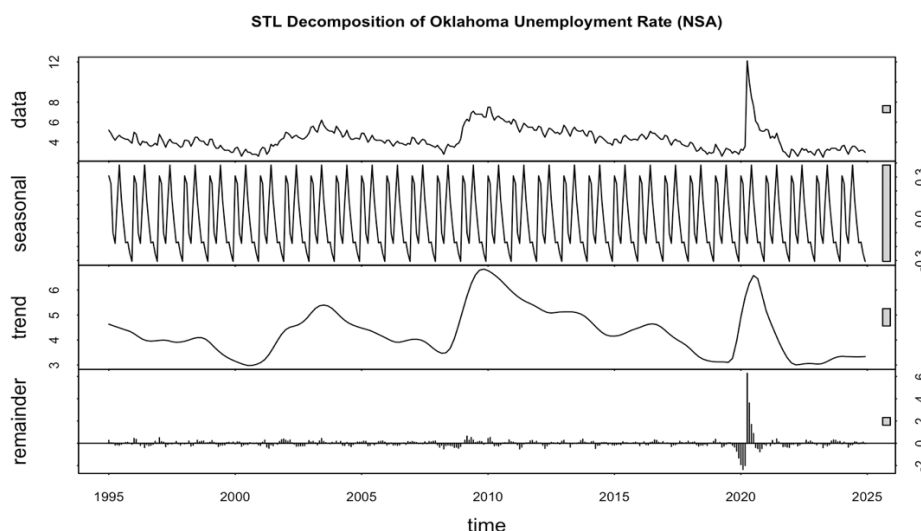
Taken as a whole, the Texas results are fairly coherent. The raw series has enough seasonal structure that it needs a seasonal model, while the adjusted series can be handled with a simpler non-seasonal ARIMA. Both approaches point in the same broad direction: Texas unemployment is highly responsive to major macroeconomic shocks, especially the Great Recession and the pandemic, but outside those episodes the state tends to move back toward a lower and more stable range. The fact that Texas shows a longer elevated period around 2009–2012 also suggests that, although the state has strong recovery capacity, adjustment in a large labor market can still take time. Overall, Texas looks cyclical but resilient: vulnerable to big national shocks, yet not showing signs of severe near-term deterioration in the forecast period.

3.2 Oklahoma

Oklahoma is smaller than Texas, but its unemployment series shows many of the same broad turning points. That makes it useful analytically, because it lets us compare a similar regional labor market with a different scale and industrial composition. In the uploaded results, Oklahoma's unemployment rate rises in the early 2000s, climbs more clearly around the Great

Recession, spikes sharply in 2020, and then returns toward a lower recent level. So the big shocks are shared with Texas, but the absolute level and short-run projection are a bit different. Oklahoma generally enters the forecast period from a lower recent unemployment base, which matters for how the final outlook is interpreted.

The Oklahoma NSA series shows the same basic problem as the Texas raw series: it contains both cyclical movement and clear annual seasonality. The uploaded STL decomposition indicates that the original series combines a trend component, a recurring seasonal component, and a remainder that becomes unusually large during crisis periods, especially in 2020. This means the raw series cannot be treated as a simple stationary process in levels. There is too much structure in it, and that structure has both seasonal and non-seasonal parts.

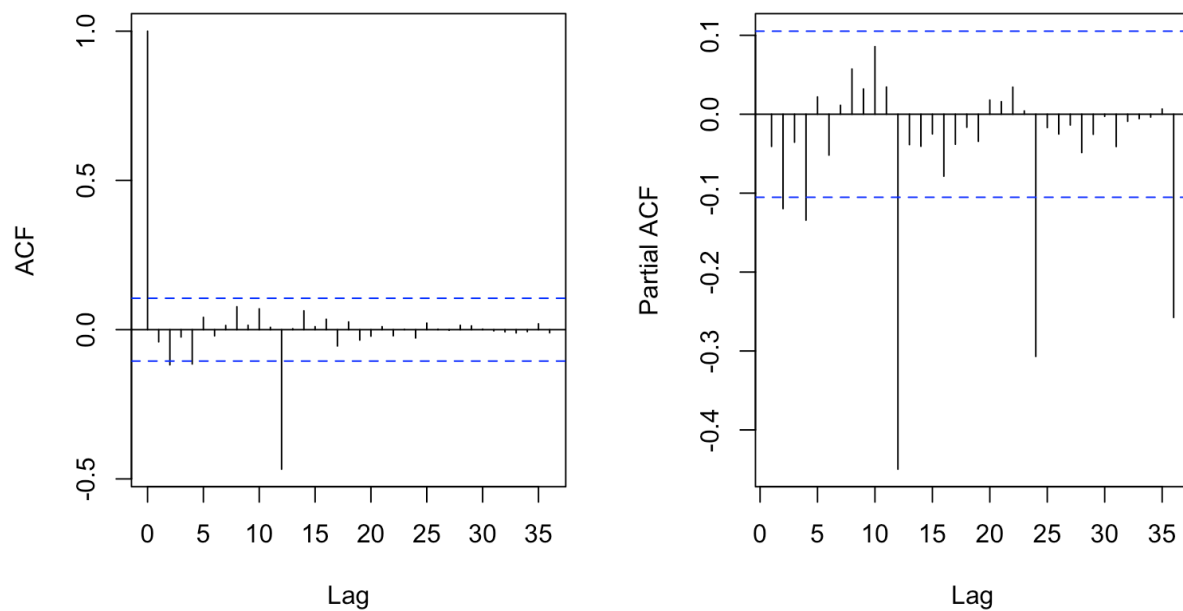


The differencing results again show why a seasonal model is needed. Seasonal differencing alone is not enough, because the broader persistence in the series remains. Ordinary first differencing alone is also not enough, because annual seasonal dependence is still visible. Only after both transformations are applied does the working series appear stable enough for low-order seasonal

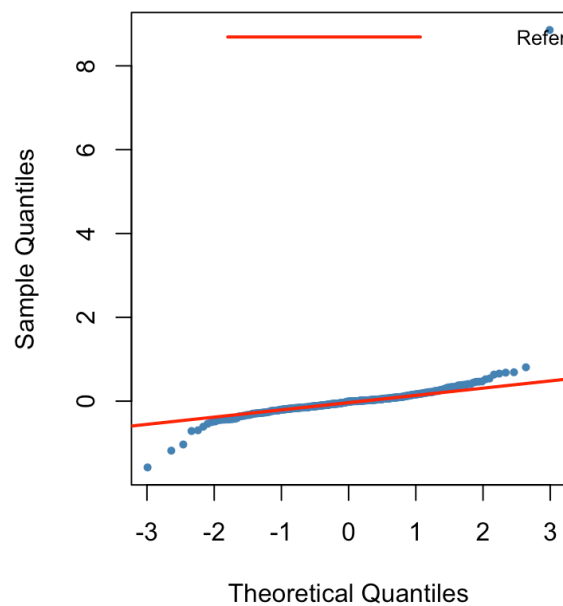
ARIMA modeling. The uploaded ACF and PACF diagnostics support

SARIMA(1,1,1)(0,1,1)[12] for the Oklahoma NSA data, which is the same broad model class used for Texas. That parallel is useful: it suggests that the two states share a similar time-series structure in their raw unemployment data, even though the level of unemployment differs somewhat between them.

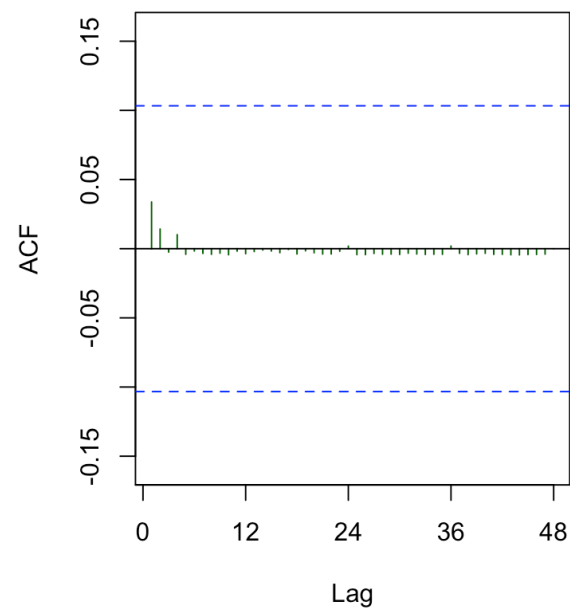
Oklahoma NSA First + Seasonal Difference:klahoma NSA First + Seasonal Difference:



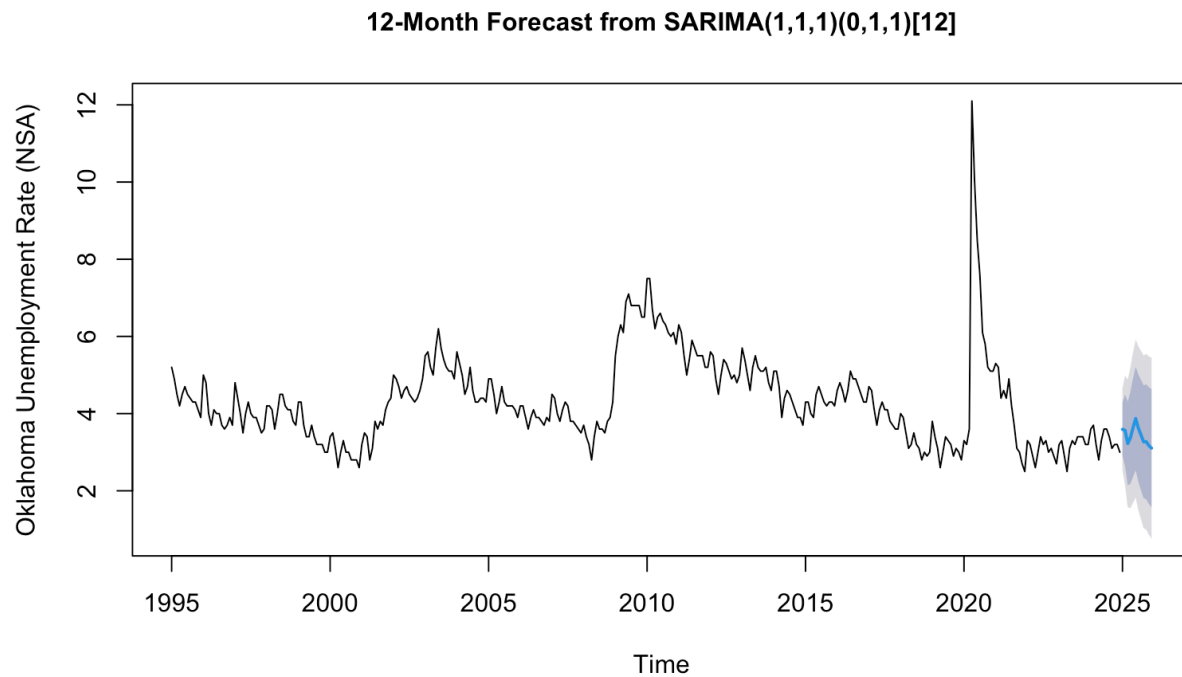
5. Q-Q Plot of Oklahoma NSA Residuals



6. ACF of Residuals² (ARCH check)

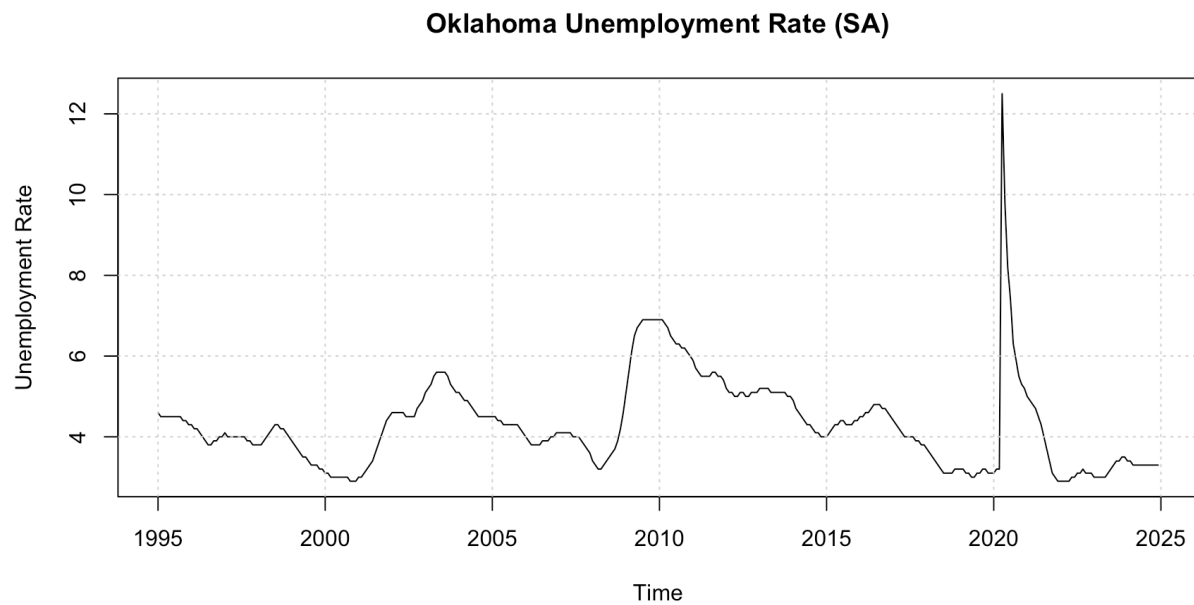


The residual evidence for Oklahoma is also fairly reasonable. Most residual autocorrelations stay inside the confidence limits, and the residual-squared ACF does not suggest strong remaining ARCH-type behavior. The main weakness again is tail behavior. The Q-Q plot shows that the pandemic months are still behaving like extreme outliers relative to the rest of the sample. This is not really a surprise, and it should not necessarily be treated as a failure of the model. It is better interpreted as a reminder that the model is built for ordinary short-run dynamics, not for rare crisis events of exceptional size.

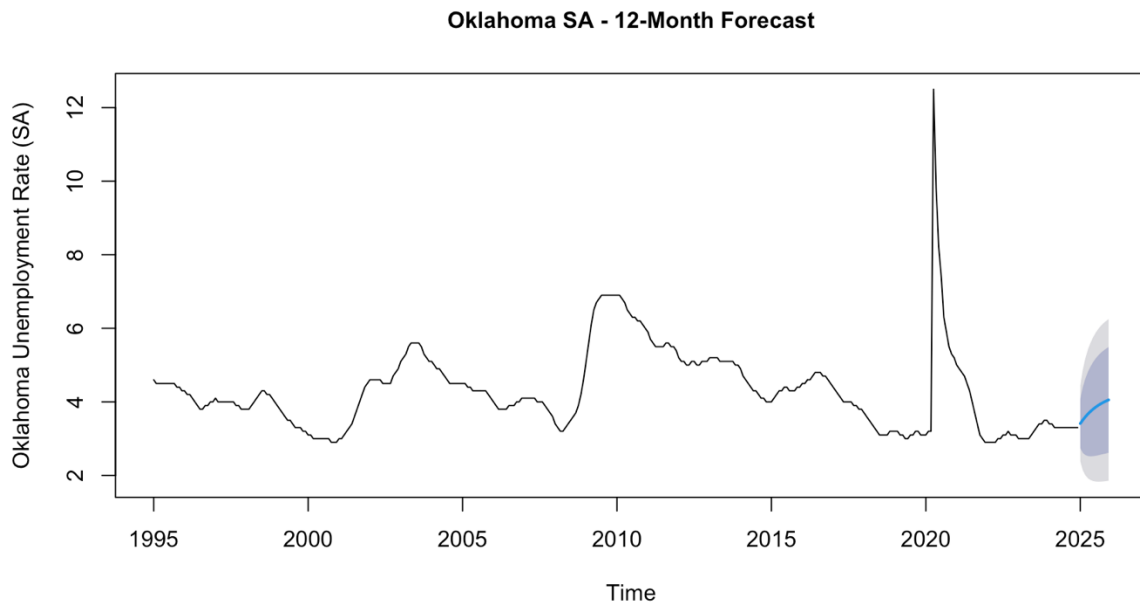


The 12-month NSA forecast keeps Oklahoma in the low- to mid-3% range. That is a little lower than Texas, and it gives Oklahoma a somewhat different near-term profile even though the two states share the same basic methodology. The forecast fan still widens with the horizon, so the model is not suggesting certainty. But the central path itself is not alarming. It implies continued stability rather than a strong new deterioration in labor-market conditions.

The Oklahoma SA series is smoother and more straightforward to read than the raw series. Once the recurring monthly pattern is removed, the cyclical path stands out more clearly: there is a rise in the early 2000s, a more visible peak around 2009–2010, the major pandemic spike in 2020, and then a return toward a lower range. In this adjusted form, Oklahoma looks like a labor market that reacts strongly to national downturns but does not remain permanently stuck at high unemployment once those shocks fade.



Methodologically, the SA results are simpler. One ordinary first difference is enough to achieve approximate stationarity, and the uploaded model outputs support ARIMA(1,1,1). As with Texas, this difference from the NSA case matters substantively. It shows that once the seasonal component has been removed in advance, the underlying unemployment dynamics can be summarized without adding seasonal terms to the model itself. That makes the SA specification easier to read and easier to compare across states.



The residual diagnostics for the Oklahoma SA model are acceptable overall. Residual autocorrelation is limited, and the ARCH check does not indicate serious volatility clustering. The main issue is again tail non-normality around the pandemic period. So, just like in Texas, the model seems suitable for normal periods but less able to absorb extraordinary crisis observations. The forecast is stable, though not completely flat. The uploaded results note a mild upward drift from a lower starting point. That is worth mentioning carefully: it does not imply a severe worsening in Oklahoma's labor market, but it does make the Oklahoma SA forecast look slightly less flat than Texas's.

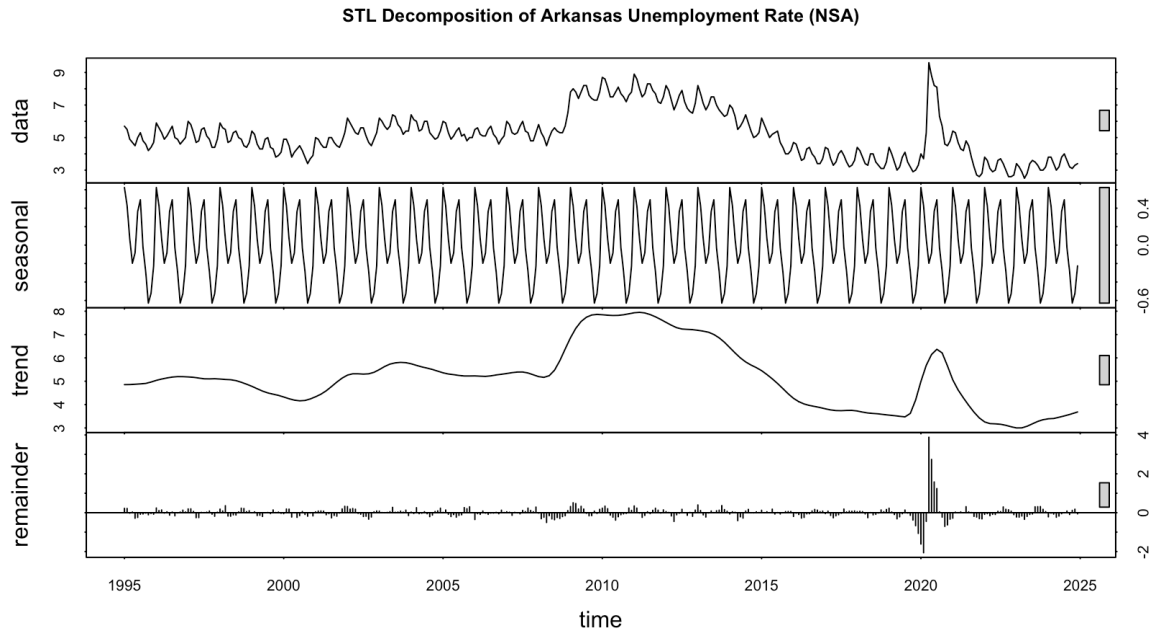
Overall, Oklahoma follows the same broad empirical logic as Texas, but with a somewhat lower recent unemployment base and a slightly softer short-run profile in the adjusted forecast. The raw data still contain enough seasonal structure to require a SARIMA model, while the adjusted data can be modeled with standard ARIMA after one ordinary difference. That consistency across NSA and SA results strengthens the interpretation. Oklahoma is clearly exposed to major

macroeconomic shocks, especially the 2020 pandemic, but the forecast evidence suggests that the state is not entering a period of severe near-term deterioration. Instead, it appears to be a smaller labor market that remains sensitive to shocks yet still relatively stable in the short run.

3.3 Arkansas

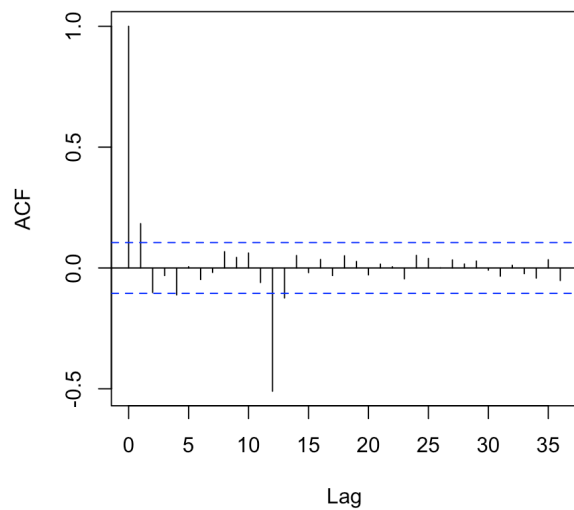
Arkansas is a useful case for state-level unemployment analysis because its labor market combines cyclical exposure with a degree of structural stability. The state economy depends on agriculture and manufacturing, but it is also supported by a strong retail and logistics base. Large employers such as Walmart, Tyson Foods, and J.B. Hunt help steady employment conditions even during downturns, which gives Arkansas a somewhat more buffered labor-market structure than states that rely more heavily on commodities or a narrower industrial base. At the same time, the unemployment series still moves clearly with major national shocks, especially the Great Recession and the COVID-19 pandemic.

The non-seasonally adjusted series shows that Arkansas unemployment contains both a business-cycle trend and a recurring within-year seasonal pattern. In the late 1990s, unemployment generally stayed in the 4% to 6% range. It then edged up during the early-2000s slowdown, rose more sharply to about 8% during the Great Recession, and gradually fell back to roughly 3.5% before the pandemic. In 2020, the unemployment rate jumped to around 9% to 10% within a short period, then returned toward pre-pandemic levels by 2022. The STL decomposition confirms that the raw series is not stationary. It contains a smooth trend, a regular seasonal component, and a remainder term that becomes extremely large during 2020.

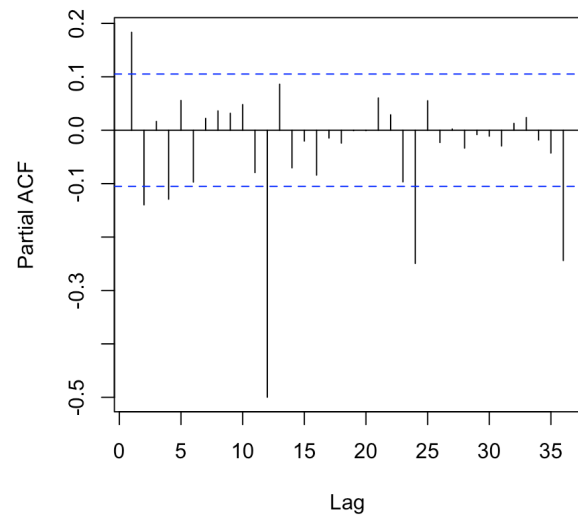


The transformation results show why a seasonal model is needed. Seasonal differencing at lag 12 removes much of the repeated annual movement, but it does not fully eliminate the underlying persistence. Ordinary first differencing removes the broad trend but still leaves noticeable annual dependence in the autocorrelation structure. Only after applying both one ordinary difference and one seasonal difference does the working series fluctuate around zero in a way that is consistent with approximate stationarity. The ACF and PACF of the transformed series then point to a low-order seasonal ARIMA specification. Among the candidate models considered, SARIMA(1,1,1)(0,1,1)[12] had the best AICc for Arkansas. Residual diagnostics are broadly acceptable: most autocorrelations remain within the confidence bounds, the Ljung–Box test does not reject the null of no remaining autocorrelation at conventional levels, and the ARCH test shows no strong evidence of volatility clustering. The main weakness is the 2020 pandemic spike, which appears as heavy tails in the Q–Q plot and drives much of the non-normality in the residuals.

Arkansas NSA First + Seasonal Difference: ACF

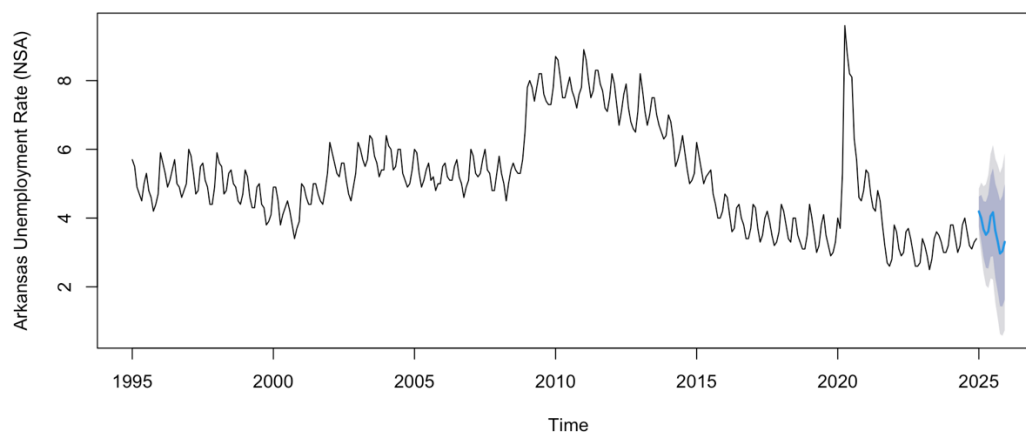


Arkansas NSA First + Seasonal Difference: PACF

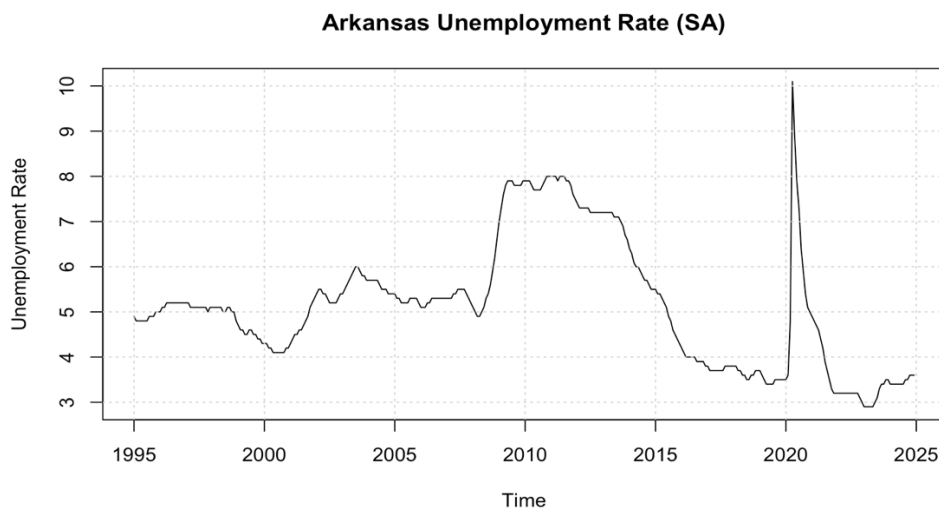


The Arkansas NSA forecast suggests a stable short-run path rather than a new deterioration. The 12-month forecast remains in the low-to-mid 3% range and preserves the seasonal oscillation that is built into the raw data. The uncertainty bands widen gradually over the horizon, which is normal in a time-series forecast and should not be interpreted as a sign of unusual instability. In substantive terms, the model suggests that Arkansas is expected to continue fluctuating within a relatively low unemployment range, with seasonality still visible in the month-to-month pattern.

12-Month Forecast from SARIMA(1,1,1)(0,1,1)[12]



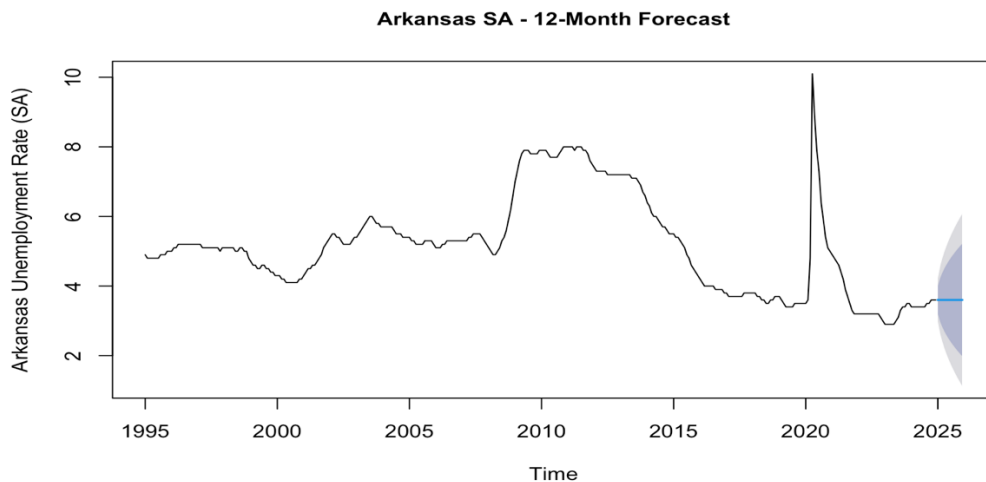
The seasonally adjusted series provides a cleaner view of the underlying labor-market cycle. Without the recurring monthly pattern, the major turning points are easier to follow: the gradual buildup during the Great Recession, the very sharp jump in 2020, and the post-pandemic recovery. In the SA data, the cyclical structure becomes more visible and easier to discuss in economic terms because the regular calendar effects no longer dominate short-run movement. This is one reason the SA series is often more useful when the goal is to describe the direction of the labor market rather than the exact seasonal pattern within a



year.

For the SA series, one ordinary first difference is sufficient to bring the data close to stationarity, and no seasonal differencing is required. The ACF decays quickly, the PACF shows limited remaining structure, and the preferred model in the new file is ARIMA(1,1,1). That is simpler than the NSA specification, which makes sense because the seasonal component has already been removed by construction. The residual results are similar to the NSA case: residual autocorrelation is limited, the ARCH test does not indicate strong conditional heteroskedasticity, and the main departure from normality again comes from the pandemic observation. The Arkansas SA forecast stays near the low-3% range and is described as roughly flat over the near

term. In other words, once seasonal noise is removed, Arkansas's unemployment path looks stable rather than sharply improving or worsening.



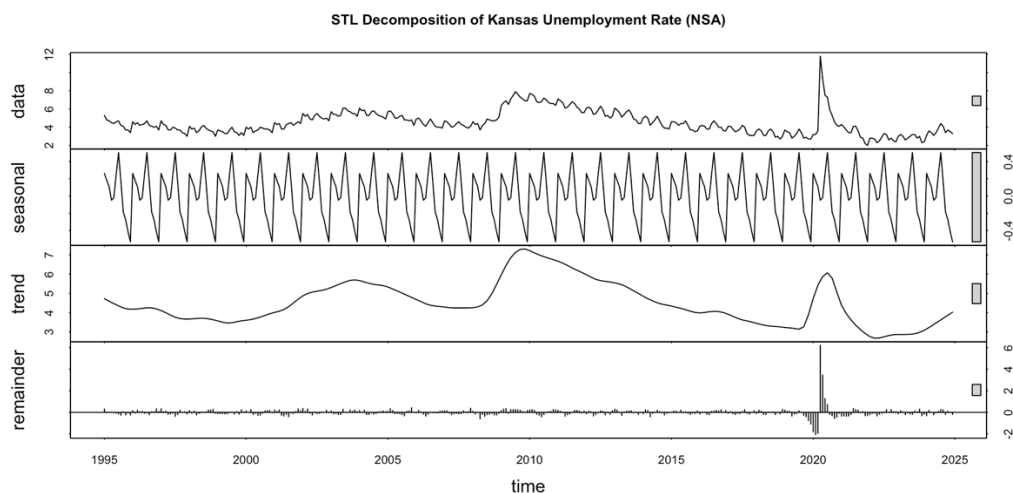
Taken together, the Arkansas results suggest a labor market that is cyclical but not unusually fragile. The state does react strongly to major national downturns, and the pandemic is by far the biggest source of model misfit, but the broader pattern is one of recovery back toward a relatively low unemployment range. Arkansas also appears to have experienced a slower buildup and a more prolonged high-unemployment phase during 2008–2012 than Kansas, which is consistent with the idea that its large retail, food-processing, and logistics employers cushion the initial shock but may also make adjustment more gradual. Looking at both NSA and SA data is especially useful here: the NSA series preserves seasonal detail, while the SA series makes the underlying trend and recovery path much easier to interpret.

3.4 Kansas

Kansas offers a useful contrast to Arkansas because it shares the same major national shocks but has a different industrial structure and a somewhat different labor-market response. Its economy

is more exposed to agriculture, aviation manufacturing, and energy extraction, all of which are sensitive to commodity prices, trade conditions, and broader shifts in demand. That means Kansas often responds more quickly and more sharply to external shocks than Arkansas. Even when the two states move through the same broad national cycle, the shape and speed of their labor-market adjustment are not identical.

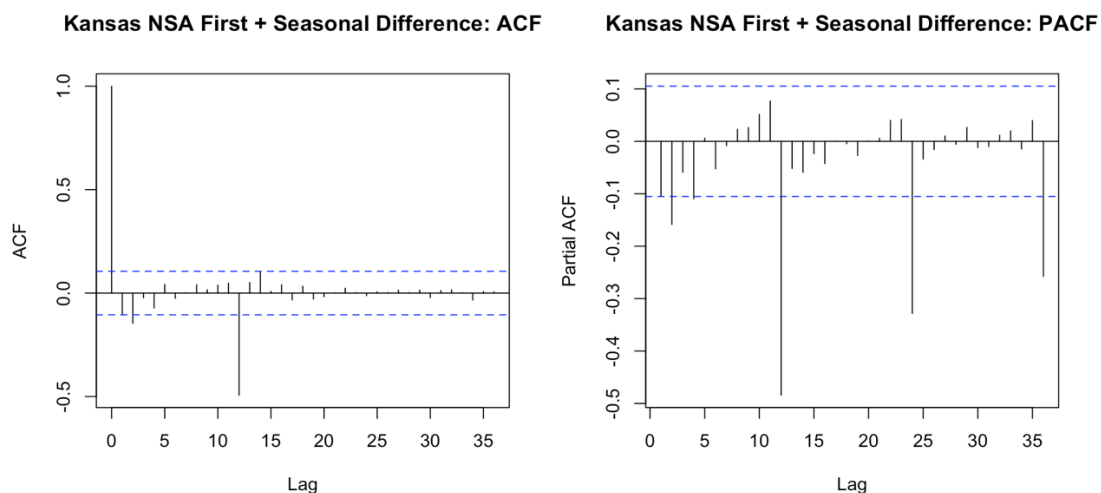
The non-seasonally adjusted Kansas series shows the same broad cyclical phases seen in Arkansas, but with some meaningful differences in magnitude. Kansas unemployment followed a similar path through the late 1990s and 2000s, but by the pre-pandemic period it had already been trending lower than Arkansas by roughly half a percentage point. During the pandemic, Kansas experienced a steeper spike, reaching close to 12%, before falling back relatively quickly. The STL decomposition shows that, like Arkansas, Kansas's raw unemployment series is not stationary in levels. It contains a smooth business-cycle trend, a recurring seasonal component, and an unusually large remainder term during



2020.

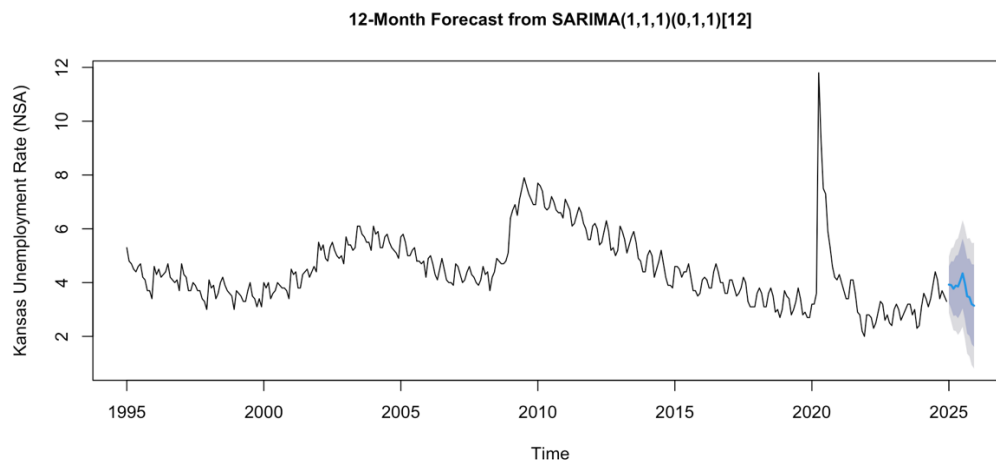
The modeling logic for Kansas NSA is very similar to Arkansas. Seasonal differencing alone does not fully remove the broader persistence in the series, while first differencing alone leaves

clear annual dependence. After combining one ordinary difference with one seasonal difference at lag 12, the working series becomes much more stable. The ACF and PACF then support a low-order seasonal ARIMA model, and the preferred specification is again SARIMA(1,1,1)(0,1,1)[12]. Residual diagnostics are broadly satisfactory: autocorrelations remain mostly within the confidence bounds, the Ljung–Box test does not indicate strong remaining serial correlation, and the ARCH test shows little evidence of strong volatility clustering. The biggest issue, as in Arkansas, is that the 2020 shock creates heavy tails in the residual distribution. Still, the model captures the ordinary month-to-month and year-to-year structure of Kansas unemployment reasonably

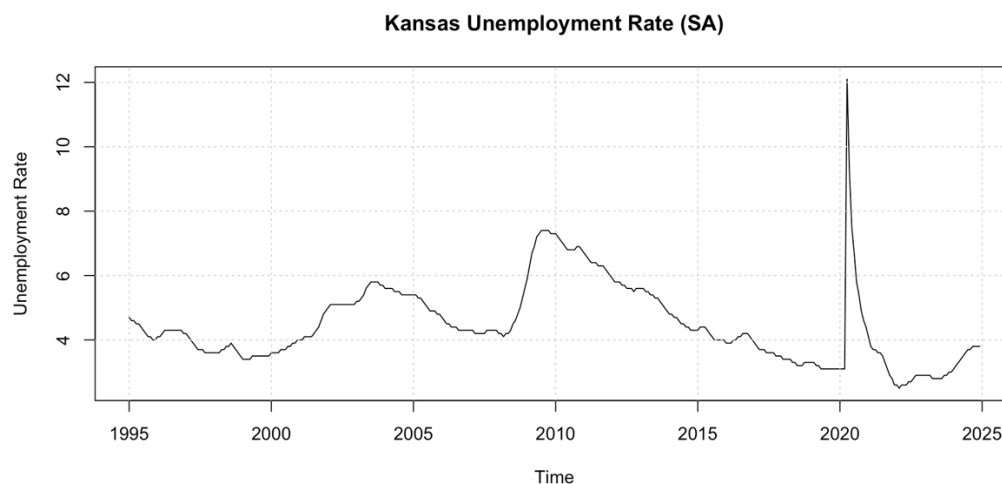


well.

The Kansas NSA forecast is also fairly stable. Over the next 12 months, the forecast remains in a relatively low range and continues to show seasonal oscillation, which is exactly what should happen when forecasting non-seasonally adjusted data. The confidence bands widen over time, but the central path does not indicate a new recession-style increase in unemployment. So even though Kansas appears somewhat more volatile around large shocks, the near-term forecast is not especially alarming.

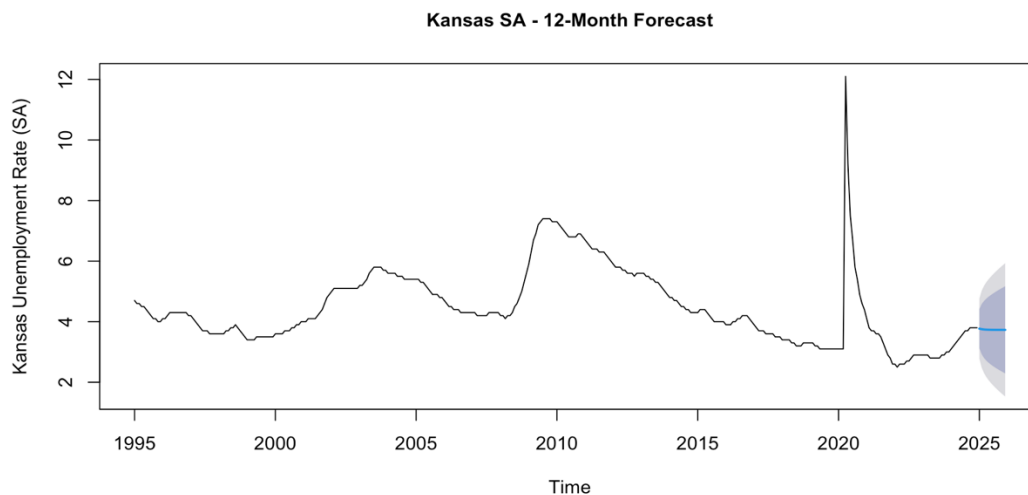


The seasonally adjusted series gives a cleaner picture of Kansas’s underlying labor-market path. Without recurring monthly variation, the series shows the main cyclical pattern more clearly: recession-related increases, a sharp pandemic spike, and then a return to a lower range. This adjusted form is better suited for judging the medium-run trend in labor-market conditions, because it removes calendar effects and makes it easier to see whether unemployment is moving up or down in a more persistent



sense.

As in Arkansas, one ordinary first difference is sufficient for the SA data, and no seasonal differencing is needed. The ACF decays quickly, the PACF shows limited additional structure, and the selected model is ARIMA(1,1,1). Residual diagnostics are broadly acceptable. The pandemic shock still produces heavier tails than a normal distribution would suggest, but outside that episode the model fit looks reasonably clean. The Kansas SA forecast remains near its recent level and, like Arkansas, does not point to notable deterioration in the near term. The outlook is stable, though uncertainty widens as the forecast extends further



out.

What makes Kansas especially interesting is not that its model class is different, but that its economic interpretation differs from Arkansas even under the same broad statistical structure. The report notes that Kansas had a sharper pandemic spike and a faster reversal. That pattern fits the state's economic makeup: a labor market tied more closely to commodity prices and manufacturing demand is likely to react more violently when conditions suddenly worsen, but it can also bounce back more quickly when those pressures reverse. In that sense, Kansas looks slightly more reactive and slightly more volatile than Arkansas, even though both states are ultimately described well by the same NSA and SA model families.

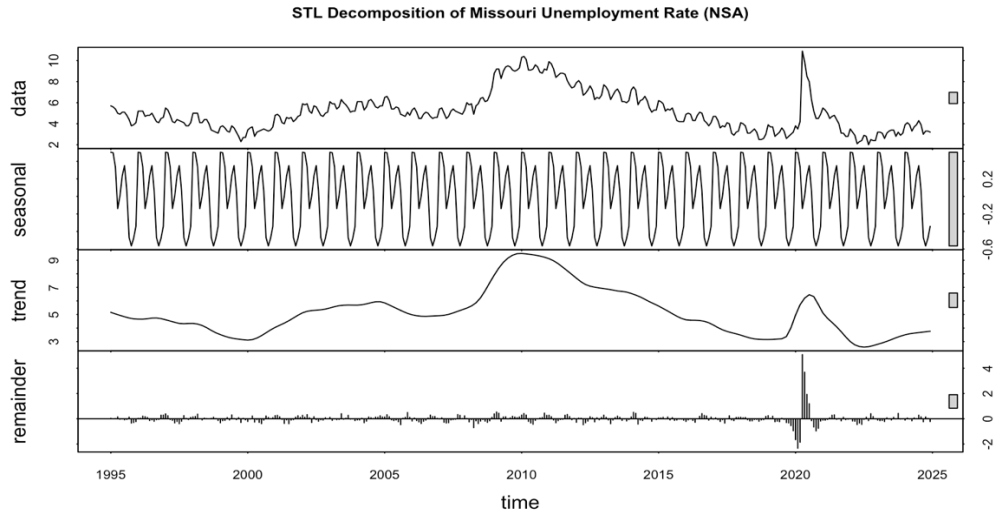
Overall, Kansas appears to be a labor market with stronger short-run responsiveness to external shocks but a still-stable near-term outlook. The NSA results show clear seasonality and justify a seasonal ARIMA specification, while the SA results isolate the underlying cyclical movement and support a simpler non-seasonal ARIMA. Like Arkansas, Kansas does not show evidence of major near-term deterioration in either forecast. The main difference is in the shape of adjustment: Kansas looks quicker and sharper around major shocks, while Arkansas looks somewhat steadier but more drawn out in recovery.

3.5 Missouri

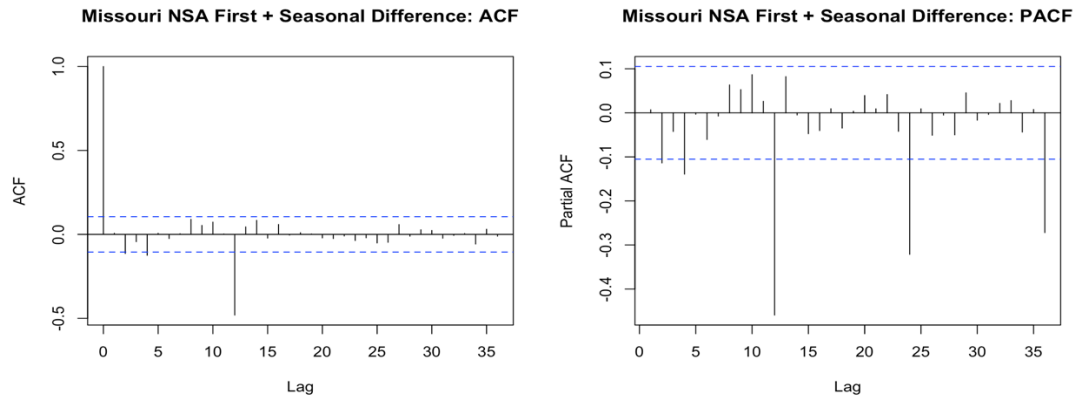
Missouri represents a labor market that is clearly cyclical, but still relatively resilient over the long run. Compared with more fragile states, Missouri does not appear to be dominated by extreme seasonal swings, although its unemployment path still reflects regular within-year movement in the raw data. This pattern fits the structure of the state economy, which combines manufacturing, retail, healthcare, agriculture, and other service-sector activity. A more diversified economic base does not eliminate exposure to national recessions, but it does appear to reduce the risk of prolonged labor-market weakness.

The non-seasonally adjusted Missouri series contains both a cyclical trend and a repeated seasonal pattern. Over the sample, the raw unemployment rate shows several major crisis peaks, including the early-1980s downturn, the 2008–2009 financial crisis, and the 2020 pandemic shock. The STL decomposition confirms that the NSA series is not stationary in levels. It contains a smooth long-run trend, a stable seasonal component, and a remainder that becomes unusually large during crisis periods, especially in 2020. This is important because it shows that the raw series contains real structure rather than just random variation. In Missouri, seasonal

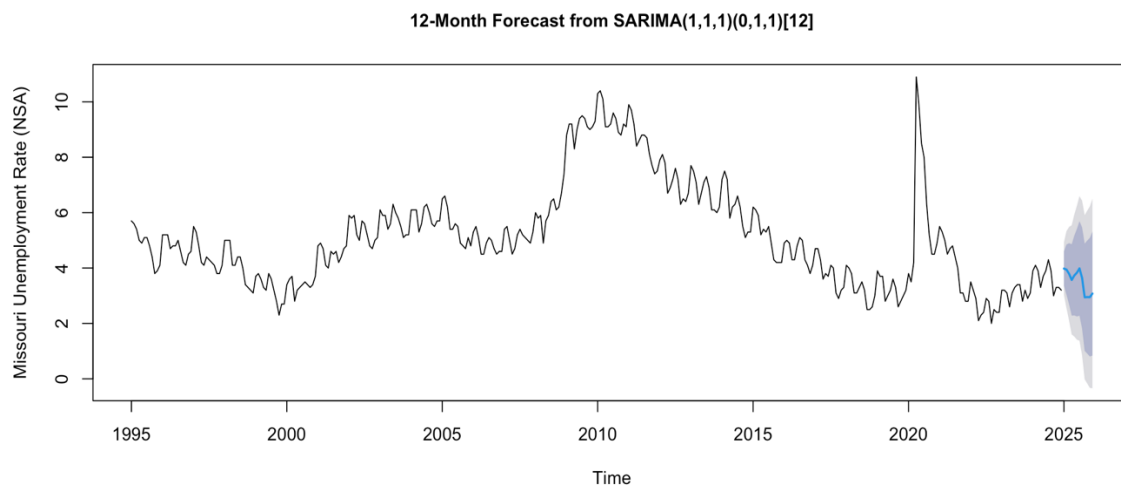
movement is present, but it is not so large that it completely dominates the broader unemployment cycle.



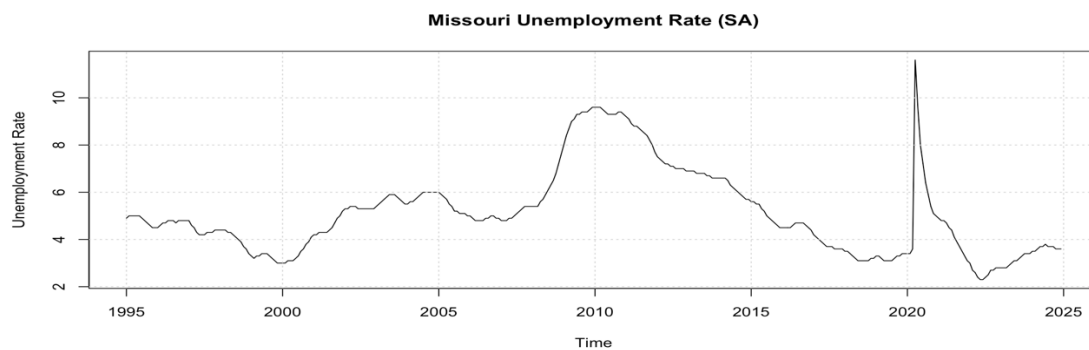
The transformation results show that seasonal modeling is needed for the raw series. Seasonal differencing at lag 12 removes much of the recurring annual pattern, but non-seasonal persistence still remains. Ordinary first differencing removes the broader trend, but leaves clear seasonal dependence in the autocorrelation structure. Only after applying both one ordinary difference and one seasonal difference does the working series fluctuate more consistently around zero. The ACF and PACF of the transformed series support a low-order seasonal ARIMA specification, and the selected model is $\text{SARIMA}(1,1,1)(0,1,1)[12]$. Residual diagnostics are acceptable overall. Most residual autocorrelations stay within the confidence bounds, and the ARCH-style check does not suggest strong remaining volatility clustering. The main limitation is the pandemic shock, which appears as a large outlier and produces heavy tails in the Q–Q plot.



The Missouri NSA forecast suggests a broadly stable short-run path. Over the next 12 months, the point forecast remains around the low- to mid-3% range and continues to display seasonal oscillation, which is exactly what should be expected in a forecast based on non-seasonally adjusted data. The forecast interval widens over time, but the central path does not indicate a major new deterioration. In other words, the model suggests that Missouri's raw unemployment rate will continue to fluctuate within a relatively normal seasonal range rather than break sharply upward.



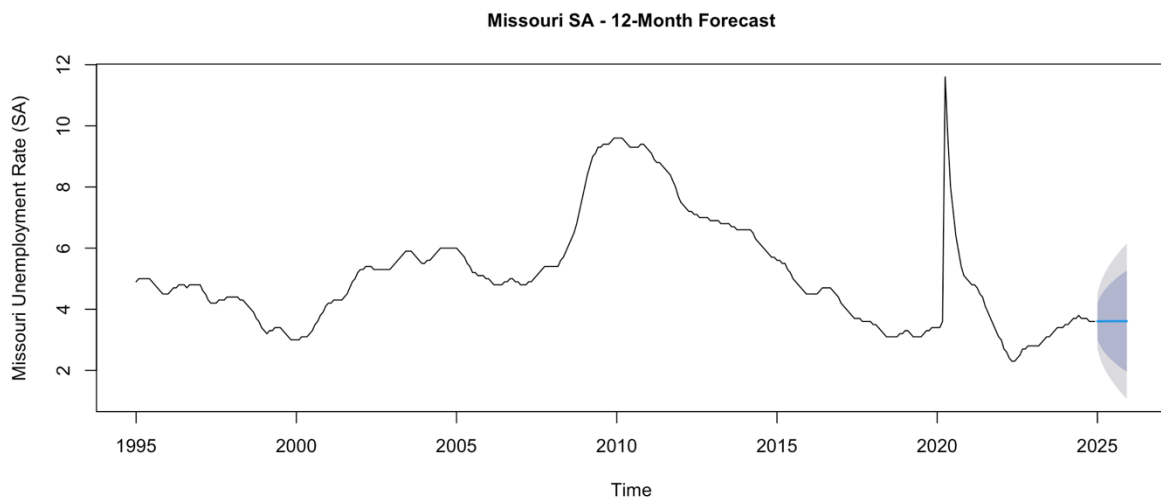
The seasonally adjusted Missouri series gives a clearer picture of the underlying labor-market cycle because recurring monthly fluctuations have already been removed. In the SA data, the main turning points are easier to interpret. Missouri unemployment stays around the mid-4% to 5% range in the mid-1990s, declines toward roughly 3% around 2000, rises again in the early 2000s, climbs to nearly 10% around 2009–2010 during the financial crisis, spikes sharply in 2020 during the pandemic, and then falls back toward the mid-3% range in recent years. This pattern is consistent with the broader conclusion in your written analysis that Missouri has experienced a long-run improvement in labor-market performance, while still remaining clearly exposed to major national downturns.



After one ordinary difference, the Missouri SA series appears approximately stationary for most of the sample. The differenced series fluctuates around zero, with the main exception being the very large shock around 2020. The ACF and PACF of the first-differenced series suggest a simple low-order ARIMA structure rather than a complicated model. Based on the uploaded figures and the written analysis, the preferred specification is ARIMA(1,1,1). That is simpler than the NSA model, which makes sense because the seasonal component has already been removed in advance. Residual diagnostics are generally satisfactory. The residual plot shows that most observations remain close to zero outside the pandemic period, and the residual ACF

indicates little remaining serial correlation. The Q–Q plot still shows non-normality in the tails, but this is driven mainly by the extreme pandemic spike rather than by a general failure of the model.

The 12-month forecast from the Missouri SA model is relatively flat and remains close to the recent unemployment level. The point forecast stays around the mid-3% range, while the confidence interval widens gradually over time. This supports the interpretation that Missouri’s labor market is not moving into a severe new downturn in the near term. Instead, after seasonal noise is removed, the short-run outlook looks stable, with normal forecast uncertainty but no evidence of a sharp worsening trend.



Taken together, the Missouri results suggest a labor market that is cyclical but not unusually fragile. The state responds clearly to major national shocks, especially the 1982 crisis, the 2008–2009 recession, and the 2020 pandemic, but it also shows meaningful recovery capacity once those shocks begin to fade. In both the NSA and SA results, the pandemic remains the largest single outlier, yet the broader pattern still points toward return to a relatively low unemployment

range. Compared with Nebraska, Missouri is more cyclical and less stable, but compared with New Mexico, it is less seasonally distorted and less vulnerable to prolonged labor-market weakness. This makes Missouri a useful middle case within the regional comparison: not the most stable state, but also not the most fragile.

3.6 Nebraska

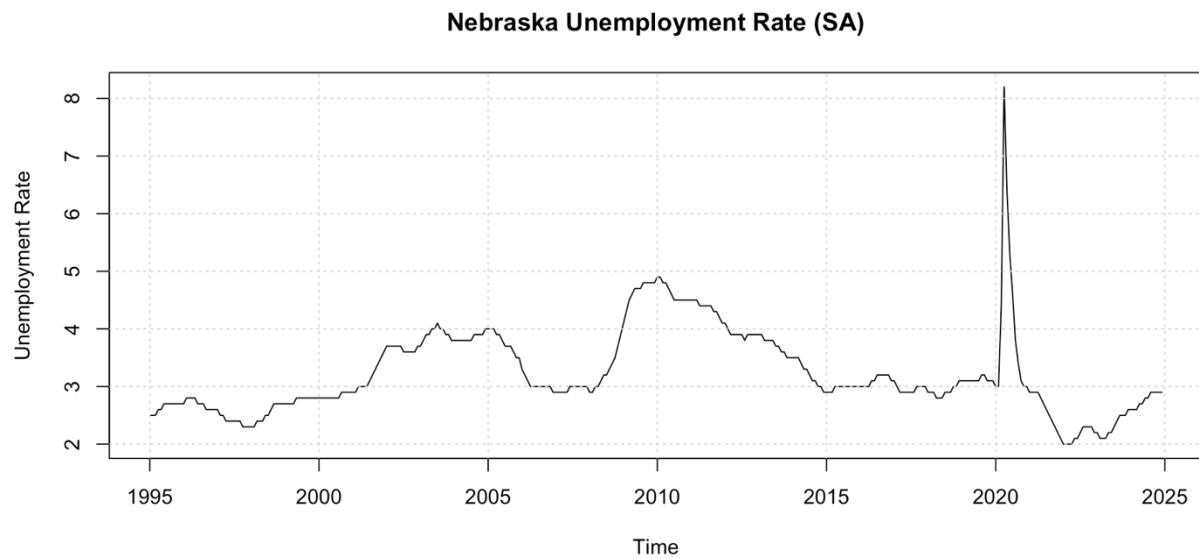
Nebraska stands out as the most stable labor market among these three states. Compared with Missouri and New Mexico, it has the lowest average unemployment rate, the narrowest peak-to-trough range, and the least visible seasonal fluctuation. This fits the structure of the state economy, which is centered on large-scale agriculture, food processing, and relatively stable non-cyclical services such as education, healthcare, and government. These industries appear to provide a strong employment base even during broader downturns, which helps explain why Nebraska behaves like an employment-stability anchor in the region.

One useful way to see this stability is to compare the non-seasonally adjusted (NSA) and seasonally adjusted (SA) series. In Nebraska, the two series stay very close to each other over most of the sample, which suggests that recurring seasonal effects are limited. According to the uploaded analysis, monthly fluctuations in the raw series rarely exceed 0.8 percentage points, which is much smaller than the variation observed in Missouri and especially New Mexico. In a stable pre-pandemic year such as 2019, the Nebraska NSA unemployment rate ranged from 1.8% in August to 2.5% in January, while the SA series remained close to 1.9%–2.1%. Even during the pandemic, the NSA–SA gap was only about 0.3 percentage points in April 2020. This suggests that seasonality exists, but it has only a limited effect on the overall labor-market path.

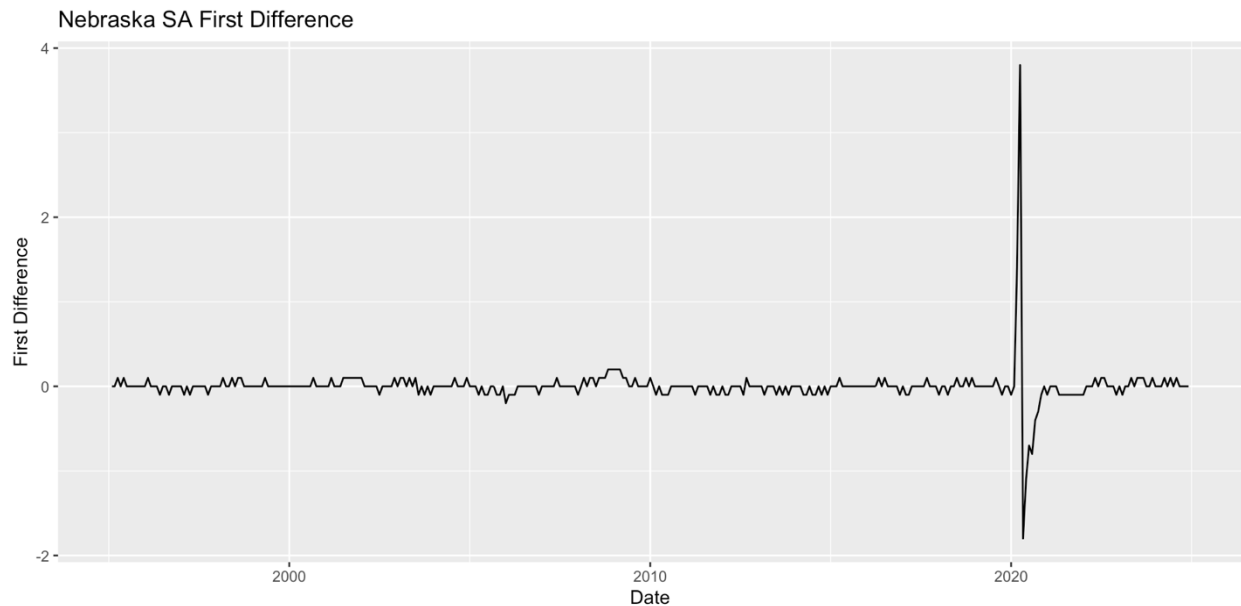
Nebraska's raw unemployment series is therefore much less affected by calendar-related movement than the raw series of more seasonal states.

This weak seasonality also has a clear economic interpretation. Nebraska's agricultural base is highly mechanized, which reduces the traditional labor swings associated with planting and harvest seasons. In addition, industries such as food processing and grain-related production tend to maintain stable demand even during broader downturns. As a result, Nebraska's labor market appears stable not only in adjusted data, but also in the raw monthly series. In this sense, the SA/NSA comparison supports the broader conclusion that Nebraska experiences very little non-crisis seasonal distortion.

The SA series makes the state's long-run stability even clearer. Nebraska's unemployment rate starts at 4.2% in January 1976, falls to 2.0% in March 2000 and 1.9% in September 2019, and ends at 3.0% in December 2024. Over the full sample, that is only a 1.2 percentage point decline, which is the smallest long-run decline among the three states. This should not be read as weak improvement. Instead, it reflects the fact that Nebraska already began from a relatively strong labor-market position and had less room to improve than a state such as New Mexico. In other words, the long-run trend suggests stability from a low base rather than lack of progress.



Another useful way to evaluate Nebraska is to compare how it responds to major crisis periods. The uploaded analysis reports three main SA peaks: 7.2% in November 1982, 5.5% in October 2009, and 8.5% in April 2020. Nebraska is the only state among the three whose SA unemployment rate never exceeds 9% during any crisis. The 2020 peak, although the highest in Nebraska's history, was still 3.2 percentage points below Missouri's 11.7% and 1.7 percentage points below New Mexico's 10.2% in the same month. Recovery was also rapid: the SA rate fell to 4.0% by October 2020 and returned to pre-pandemic levels by June 2021. This faster adjustment is consistent with Nebraska's concentration in essential industries, especially agriculture and food processing, which continued operating during the pandemic and supported labor-market stability.



Another important feature of the SA series is persistent full employment before the pandemic. The uploaded file notes that Nebraska's SA unemployment rate remained below 3.0% for 87 consecutive months from 2014 to 2020. That is a strong sign that the labor market was not simply stable, but unusually tight for a sustained period. In practical terms, Nebraska appears to be a state where labor shortages can become just as relevant as unemployment risk. This also helps explain why even large national shocks do not push the state into the same unemployment range seen elsewhere in the region.

Taken together, the Nebraska results are supported by three main types of state-level evidence: SA/NSA comparison, long-run trend analysis, and crisis peak-recovery analysis. All three point in the same direction. Nebraska is unusually stable, only mildly seasonal, and highly resilient to major shocks. Compared with Missouri, it is less cyclical and less seasonally affected. Compared with New Mexico, it is far less volatile and much quicker to recover. In the broader regional comparison, Nebraska clearly represents the low-volatility end of the spectrum and serves as the strongest example of labor-market stability in this report.

3.7 New Mexico

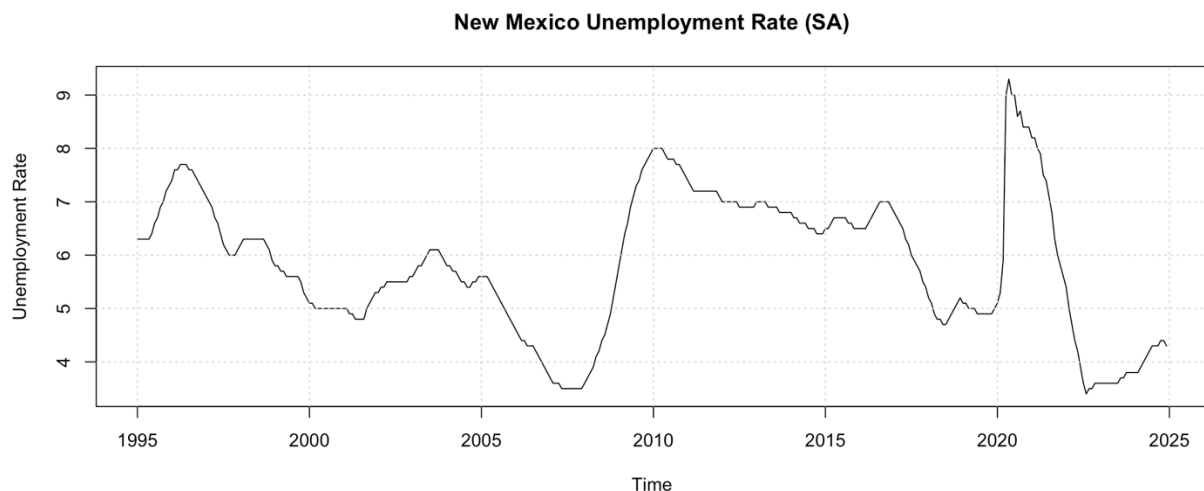
New Mexico is the most sensitive labor market among these three states. Compared with Missouri and Nebraska, it starts from a higher unemployment level, shows stronger seasonal movement, and tends to recover more slowly after major shocks. This pattern is consistent with the structure of the state economy. Historically, New Mexico has depended more heavily on energy extraction, tourism, and government-related activity, which makes the labor market more exposed to both cyclical downturns and recurring within-year fluctuations. In the broader regional comparison, New Mexico represents the high-volatility end of the spectrum.

One useful way to see this sensitivity is to compare the non-seasonally adjusted (NSA) and seasonally adjusted (SA) series. In New Mexico, the gap between the two is much larger than in Missouri and Nebraska, which suggests that seasonal effects are an important part of the unemployment pattern rather than a minor statistical adjustment. According to the uploaded analysis, monthly fluctuations in the raw series are generally in the 1.5 to 2.5 percentage point range, which is the largest among the three states. In a relatively normal year such as 2019, the NSA unemployment rate rises from 3.8% in July to 5.2% in January 2020, while the SA rate remains much more stable near 3.8% to 4.0%. This suggests that recurring within-year movement is a meaningful feature of the labor market and needs to be taken seriously when interpreting the raw unemployment series.

The economic explanation for this stronger seasonality is also clear. Tourism activity raises labor demand during stronger travel periods and lowers unemployment temporarily relative to the adjusted series, while off-season slowdowns push unemployment back up. Energy-related sectors also contribute to instability because hiring and layoffs can react to shifts in prices and

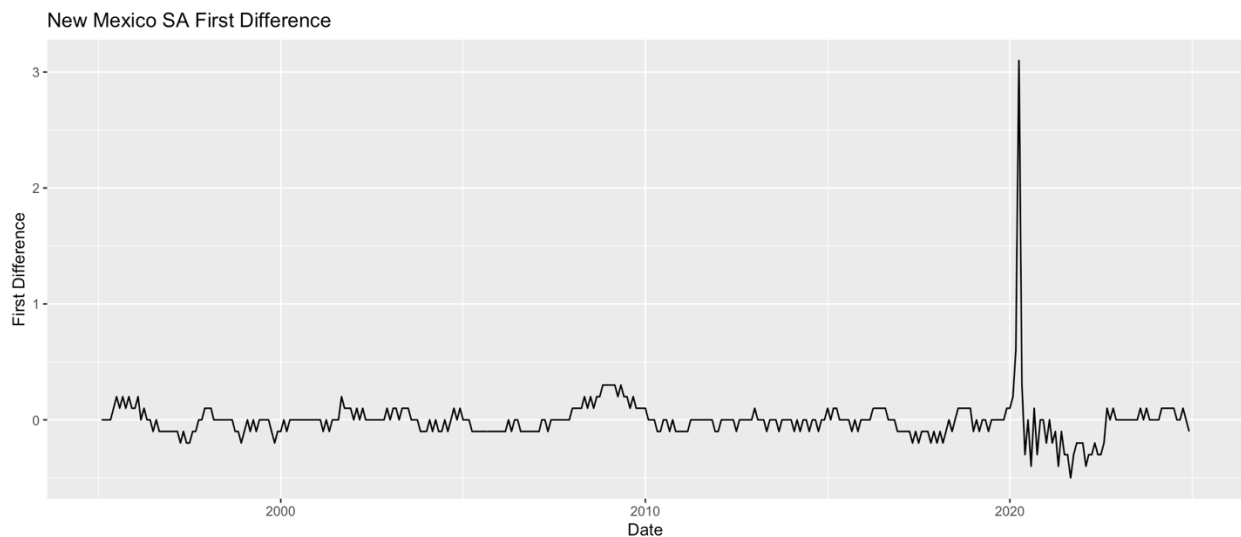
production conditions. As a result, the raw monthly unemployment series in New Mexico looks less smooth and more sensitive than in the other two states. In practical terms, the SA/NSA comparison shows that both seasonal effects and broader cyclical shocks matter for understanding labor-market conditions in the state.

The SA series gives a clearer picture of the underlying long-run path once recurring monthly effects are removed. In the adjusted data, New Mexico starts with the highest unemployment rate among the three states, at 9.2% in January 1976, and falls to 3.9% in February 2020 before the pandemic. By December 2024, the SA unemployment rate stands at 4.5%. Over the full sample, this is a cumulative decline of 4.7 percentage points, which is the largest long-run decline among Missouri, Nebraska, and New Mexico. Although New Mexico appears more volatile in the short and medium run, the long-run trend still suggests clear improvement over time. In that sense, the state combines short-run sensitivity with long-run catch-up.



Another useful way to evaluate New Mexico is to compare how it responds to major crisis periods. Its most severe historical peak is not the pandemic, but the early-1980s energy crisis.

The uploaded analysis reports an SA unemployment peak of 11.8% in December 1982, compared with 10.2% in April 2020. This is important because it reflects how strongly the state was tied to energy extraction in the earlier period. By 2020, the economy had become somewhat more diversified, which reduced the relative size of the unemployment spike even during a combined pandemic and oil-price shock. Even so, recovery remains slower than in Missouri and especially Nebraska. After the 2020 peak, the SA rate falls to 5.0% by April 2021 but does not return to pre-pandemic levels until July 2022. The uploaded analysis links this slower adjustment partly to the importance of tourism, which was hit hard during the pandemic and took longer to recover.



Another important point is that volatility and long-run improvement exist at the same time. New Mexico is not simply a weak labor market. Instead, it is a state with stronger exposure to shocks, but also with evidence of long-run catch-up. The file suggests that this longer-run decline in unemployment is consistent with gradual diversification away from extreme dependence on volatile energy markets and toward federal employment, renewable energy, and advanced

manufacturing. That interpretation helps explain why New Mexico can look both highly unstable in the short run and improved in the long run.

Taken together, the New Mexico results are supported by three main types of state-level evidence: SA/NSA comparison, long-run trend analysis, and crisis peak-recovery analysis. All three point in the same direction. New Mexico is both the most seasonally sensitive and the most shock-sensitive labor market among the three states. At the same time, it also shows the strongest long-run decline in unemployment, which means the story is not only one of weakness, but also of gradual improvement over time. Compared with Nebraska, New Mexico is far more volatile and much less stable. Compared with Missouri, it is more seasonal and slower to recover after major downturns. In the logic of the report, New Mexico serves as the clearest example of a labor market where both cyclical sensitivity and seasonal structure are too important to ignore.

4. Cross-State Analysis

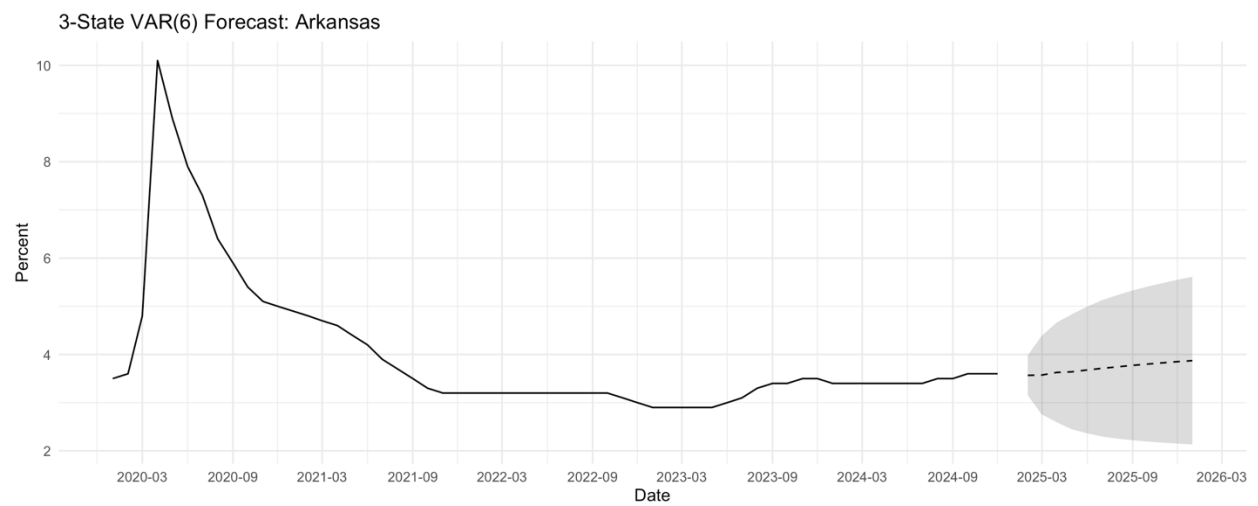
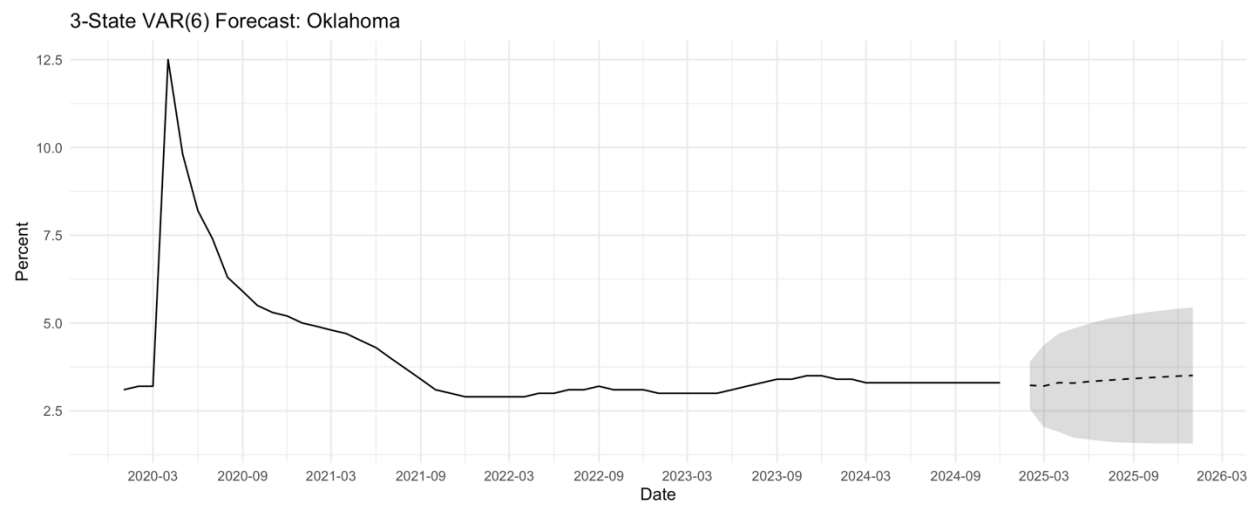
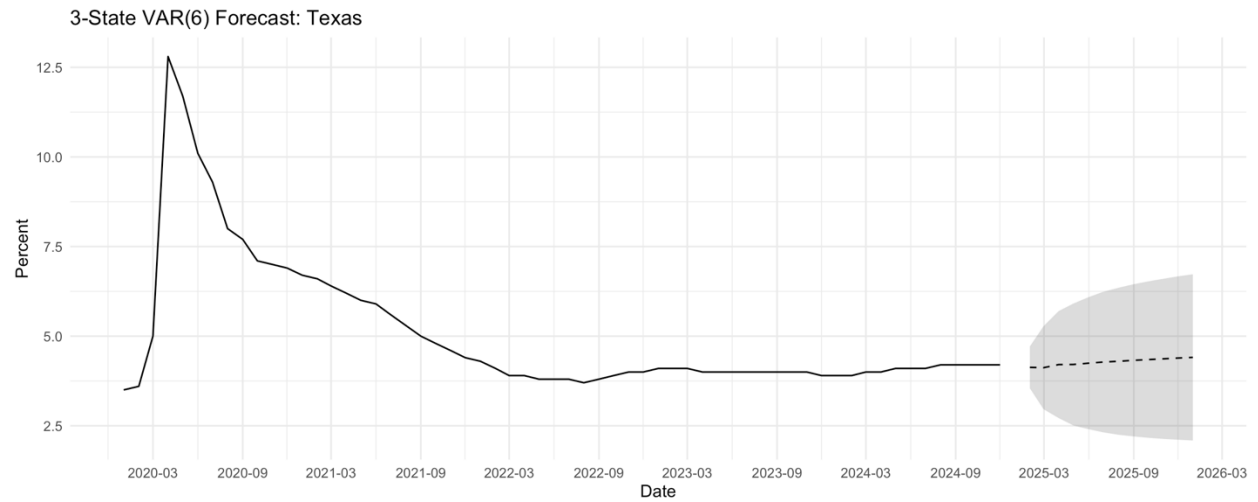
While the single-state analysis is useful for identifying persistence, seasonality, and short-run forecasting patterns within each state, it cannot show whether unemployment shocks spread across state borders. A cross-state framework is therefore needed. In this section, vector autoregression (VAR), impulse response functions (IRFs), forecast error variance decomposition (FEVD), and forecasting are used to study interstate dependence in regional unemployment dynamics. The analysis begins with a seven-state VAR(5) model to capture broader comovement across the South-Central region. Because a system with seven endogenous variables is not especially easy to interpret in detail, the discussion then turns to a more focused three-state VAR(6) for Texas, Oklahoma, and Arkansas. This smaller system makes the main transmission

channels easier to read and provides a clearer picture of how shocks move across neighboring labor markets.

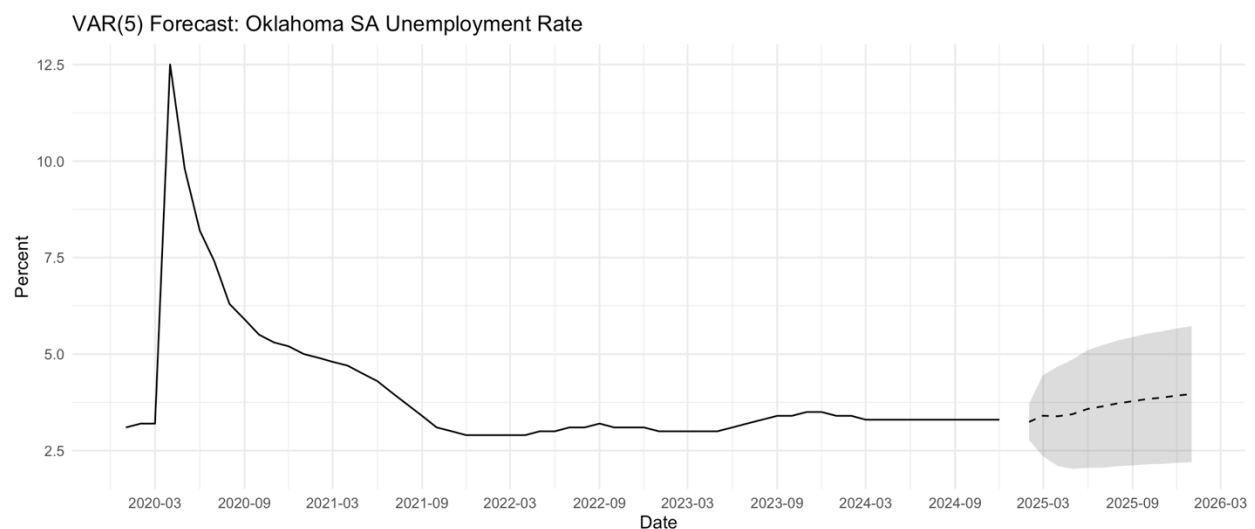
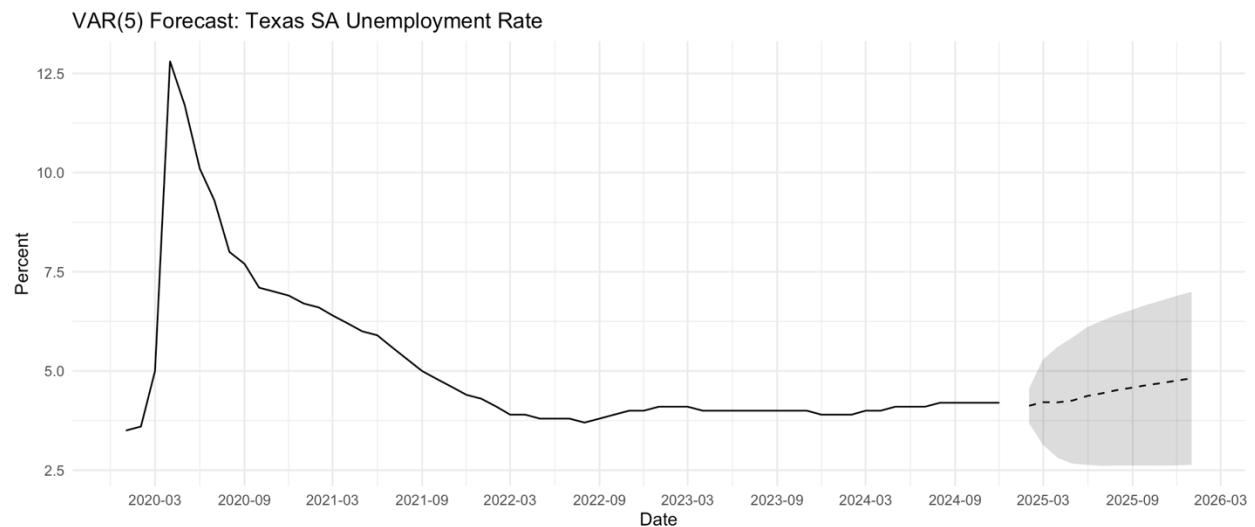
The choice of Texas, Oklahoma, and Arkansas is economically sensible. Texas is the largest labor market in the sample and plays a central role in the region. Oklahoma is closely tied to Texas geographically and economically, while Arkansas provides another useful comparison within the same broader regional setting. Looking at these three states together makes it possible to study both regional linkage and differences in how strongly each state reacts to outside shocks.

Forecast Results from the Three-State and Seven-State Systems

The three-state VAR(6) forecasts indicate mild upward movement in unemployment over the forecast horizon, but they do not suggest a sharp deterioration in labor-market conditions. For Texas, the point forecast rises from just above 4.1% to about 4.4% by early 2026. Oklahoma follows a similar path, increasing from around 3.3% to roughly 3.5%, while Arkansas trends upward more modestly from about 3.6% to nearly 3.9%. The confidence bands widen noticeably over time, so the medium-run forecasts become less precise as the horizon expands. The key point is not the exact increase in each state, but the fact that none of the three projected paths shows a sudden break from the recent pattern. Instead, the model suggests relative stability with some moderate upward pressure.

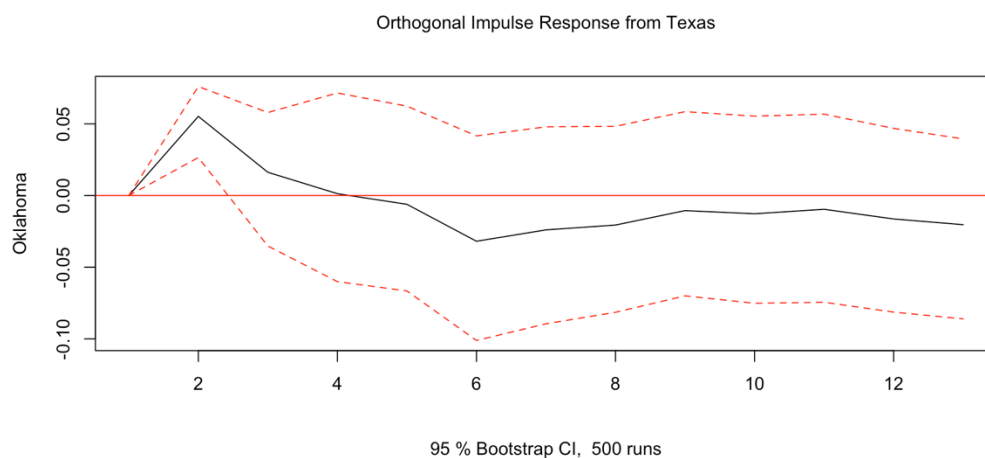


The seven-state VAR(5) forecasts support the same general interpretation. Even once the broader regional system is taken into account, the projected paths for Texas and Oklahoma remain relatively stable, although both drift upward modestly. This is useful because it shows that the moderate forecast pattern is not driven only by the smaller three-state subsystem. Instead, it appears in both the narrow and broad regional specifications. Taken together, the forecasting evidence suggests that near-term weakening is possible, but the model does not indicate a severe unemployment shock in the region.

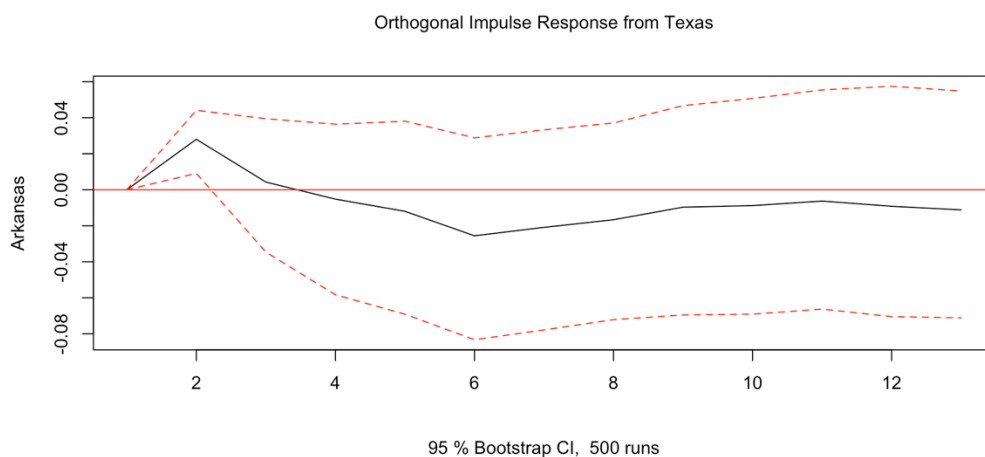


Impulse Response Evidence on Interstate Spillovers

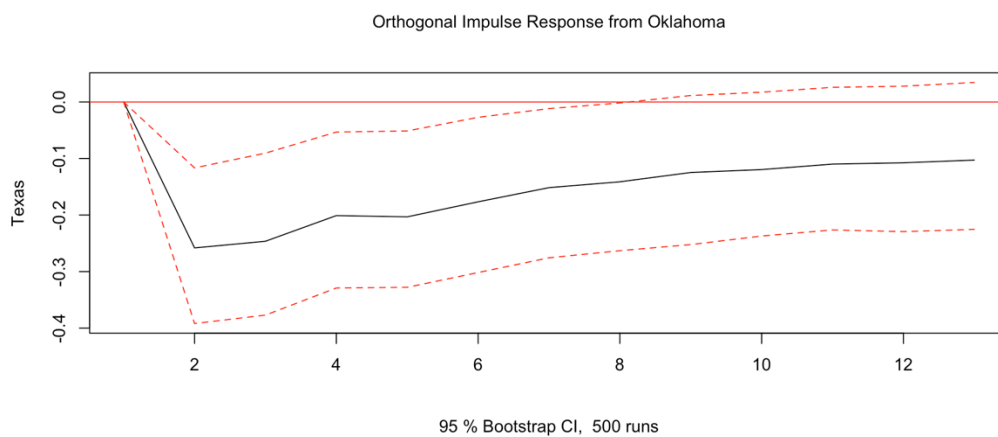
The impulse response results provide more direct evidence on interstate spillovers. Within the three-state VAR, a shock to Texas produces a positive response in Oklahoma. The response is strongest in the early periods and then declines gradually over time. This suggests that Texas acts as an important source of labor-market transmission within the subsystem rather than simply moving passively with the rest of the region. The effect is not permanent, but it is economically meaningful in the short run.



A similar pattern appears in the response of Arkansas. In the three-state VAR, a shock from Texas also raises Arkansas unemployment in the short run. Here again, the largest effect occurs early, after which the response becomes smaller over the remaining horizons. When read together, the Texas→Oklahoma and Texas→Arkansas IRFs suggest that Texas plays the leading transmitting role in this three-state system. In other words, Texas appears to be the main source state through which unemployment disturbances spread outward to the smaller neighboring labor markets.



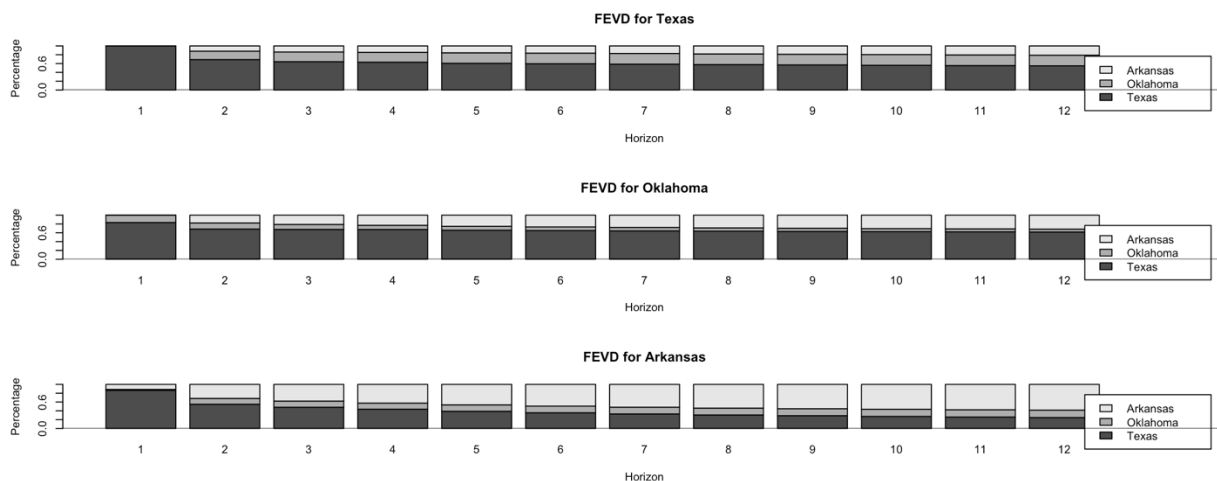
The reverse relationship is noticeably weaker. When the shock originates in Oklahoma, the response of Texas is negative for much of the horizon and only gradually returns toward zero. This matters because it shows that the spillover relationship is not symmetric. The influence running from Texas to Oklahoma is stronger than the influence running in the opposite direction. That asymmetry is one of the most important results in the cross-state section because it suggests that regional unemployment dynamics are not simply bidirectional in equal strength. Instead, the larger regional labor market appears to transmit more than it receives.



The broader seven-state IRFs are weaker and visually less pronounced than the three-state results. One likely reason is that, once the full regional system is included, part of each bilateral relationship is absorbed into wider common dynamics. For that reason, the three-state VAR is more useful for illustrating the main transmission channels, while the seven-state VAR is more useful for showing that broader regional interdependence exists at all.

FEVD Results and Core Findings

The FEVD results point in the same direction as the IRFs. For Texas, most of the forecast error variance is still explained by its own shocks, which means that internal dynamics remain the dominant source of unemployment variation. At the same time, shocks from Oklahoma and Arkansas account for a nontrivial share as the horizon expands. For Oklahoma, a relatively large portion of forecast variance is explained by shocks from Texas, which suggests stronger exposure to the regional core state. For Arkansas, outside shocks also become increasingly important over time, especially those coming from Texas and Oklahoma. This means that Oklahoma and Arkansas are not driven only by their own internal labor-market conditions; they also absorb a meaningful share of disturbances generated elsewhere in the subsystem.



The FEVD evidence is important because it tempers the IRF interpretation in a useful way. The IRFs show the direction and sign of the transmission mechanism, while the FEVD shows that internal persistence still matters a great deal. Texas remains mostly self-driven, but cross-state shocks explain a meaningful share of forecast variance over time, especially for Oklahoma and Arkansas. So the most reasonable conclusion is not that these labor markets are completely dominated by interstate spillovers, but rather that within-state persistence and interstate dependence operate together.

Overall, three findings stand out. First, unemployment dynamics in this region are not purely state-specific; there is clear evidence of cross-state linkage. Second, within the three-state subsystem, Texas appears to be the most influential transmitting state. Third, the estimated spillover effects are strongest in the short run and become weaker over time. In summary, the seven-state VAR(5) establishes the presence of broader regional comovement, while the Texas–Oklahoma–Arkansas three-state VAR(6) provides a more transparent view of how shocks are transmitted across states. Taken together, the forecast, IRF, and FEVD results suggest that unemployment dynamics in the South-Central region are shaped by both within-state persistence and interstate transmission.

5. Discussion

5.1 Why SA and NSA Both Matter

One of the clearest findings of this report is that both non-seasonally adjusted (NSA) and seasonally adjusted (SA) unemployment series are useful, but they serve different purposes. The NSA series preserve recurring within-year variation, which is important because seasonal

movement in unemployment is not just statistical noise. In several states, especially those with stronger exposure to agriculture, logistics, tourism, or other seasonally sensitive sectors, the raw data retain meaningful information about how labor demand changes across the year. For that reason, the NSA series are valuable for understanding the full observed pattern of unemployment and for identifying whether seasonal ARIMA models are needed.

At the same time, the SA series provide a clearer view of the underlying labor-market cycle. Once recurring monthly fluctuations are removed, it becomes easier to interpret medium-run movements associated with recession, recovery, and structural adjustment. The SA results also show why the adjusted data are often easier to model: in most cases, a single ordinary difference is enough, and the preferred specifications are simpler than in the NSA case. Taken together, the comparison between NSA and SA data strengthens the analysis. The NSA series preserve economically meaningful seasonal structure, while the SA series isolate the broader cyclical trend. Using both therefore produces a fuller interpretation of unemployment dynamics than relying on only one version of the data.

5.2 Why States Differ

Although the seven states are located in the same broad region and are exposed to many of the same national shocks, their unemployment patterns are not identical. This is not surprising, because industrial structure, state size, and sector concentration differ substantially across the sample. Larger and more diversified labor markets, especially Texas, are likely to absorb shocks differently from smaller or more specialized states. A state with broad exposure to services, logistics, manufacturing, and technology may have more internal offsetting forces than a state that depends more heavily on agriculture, energy, tourism, or a narrower set of industries.

The results across the states are consistent with this interpretation. Nebraska appears relatively stable, while New Mexico looks more sensitive and more seasonal. Arkansas seems steadier but more drawn out in recovery, whereas Kansas appears somewhat sharper and more reactive around major shocks. Oklahoma and Texas share similar cyclical turning points, but they differ in scale and in the level of recent unemployment. These differences suggest that unemployment should not be interpreted as a purely regional average outcome. Even when the same national event affects all states, the timing, magnitude, and persistence of the response can differ depending on the structure of the state economy. In that sense, time-series results are easier to understand when they are linked back to the economic composition of each labor market.

5.3 Why Cross-State Linkage Matters

The cross-state analysis adds another important layer to the interpretation. Single-state models are useful for describing each state's own persistence, seasonal structure, and short-run outlook, but they cannot show whether unemployment disturbances spill over across state borders. The VAR, IRF, and FEVD results suggest that such spillovers do exist. In particular, the three-state system indicates that Texas plays a more influential transmitting role, while Oklahoma and Arkansas appear more exposed to outside shocks. This means that unemployment in the region is not driven only by internal state conditions. It is also shaped by interstate relationships.

This finding matters because it changes how the regional labor market should be understood. If unemployment shocks can spread through trade links, transportation networks, production chains, or shared business conditions, then a state's unemployment path cannot be interpreted as fully independent of its neighbors. At the same time, the FEVD results also show that own-state dynamics still remain important. So the most reasonable conclusion is not that regional spillovers

dominate everything, but that state-specific persistence and interstate transmission operate together. This is exactly why the report benefits from combining single-state and cross-state methods. Looking only within states would miss regional linkage, while looking only across states would miss the substantial differences in structure and adjustment at the state level.

6. Conclusion

This report studied unemployment dynamics in seven South-Central states: Texas, Oklahoma, Arkansas, Kansas, Missouri, Nebraska, and New Mexico. To do this, it combined single-state and cross-state time-series analysis. At the state level, both non-seasonally adjusted (NSA) and seasonally adjusted (SA) unemployment series were examined using decomposition, differencing, ACF/PACF, ARIMA or SARIMA models, and forecasting. One clear result is that the NSA series usually keep important seasonal patterns, while the SA series make the underlying trend easier to see. The state-level results also show that these seven states do not behave in exactly the same way. Some appear more stable, while others are more sensitive to large shocks such as the Great Recession and the COVID-19 pandemic.

The cross-state results add another side to the analysis. They show that unemployment should not be treated as completely separate across states, since some spillover effects can be seen in the VAR, IRF, and FEVD results. This is especially clear in the Texas–Oklahoma–Arkansas system, where Texas seems to have a stronger influence on nearby states than the other way around. Overall, using both single-state and cross-state methods gives a more complete picture than using only one of them. The short-run forecast results are also fairly stable for most states and for the region as a whole. Although forecast uncertainty becomes wider over time, the results do not suggest that the region is entering a severe new unemployment shock in the near future.

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